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THE EFFECT OF ROOM ACOUSTICS ON SUBWOOFER PERFORMANCE AND LEVEL SETTING

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Abstract - Subwoofers are used as a part of both 2-channel and multichannel domestic sound reproduction systems, both with and without bass redirection. Room effects and a lack of convenient means for setting the subwoofer level make system setup difficult. The objective and subjective effects on subwoofer performance are evaluated. The room acoustic transfer function was measured for eight rooms and the amount of low frequency acoustic gain and loudspeaker interaction was calculated. Subjective experiments were conducted to assess the effect of low frequency room acoustics by applying the measured room impulse response to program material reproduced over headphones. Subwoofer level calibration using both program material and test signals was studied and specific recommendations are made.

I. INTRODUCTION

Subwoofers have been an important addition to 2-channel audio reproduction systems for many years. The introduction of multichannel digital distribution systems and the growing popularity of home theatre have caused a huge increase in the use of subwoofers. In some cases the subwoofer is called upon to reproduce only the output of the LFE (Low Frequency Enhancement) channel, but in a majority of cases it also reproduces the bass for some or all of the main channels. In setting up such a system, there are a number of options involving the relative amounts of bass from each of the main channels, the placement of the subwoofer in the room, and the actual level setting of the subwoofer. All of these will affect the overall perceived result.

It is well known that the room response for frequencies below the Schroeder frequency is extremely non-uniform. In particular, below about 200 Hz in average sized domestic rooms, the room modes which transfer energy from the speaker to the listener are sparsely distributed. That sparse distribution, and the fact that the coupling of both the speaker and the listener to the room modes depends upon location, contribute to the irregular response.

How should the user set the level of the subwoofer? The measured and perceived response can be so irregular as to make it very difficult to characterize the low frequency response in order to make a judgement as to the appropriate subwoofer level.

We seek to answer the following questions:

- What is the range of subwoofer transfer functions expected in small rooms?
- Is the perceived level best characterized by the maxima, average level, or minima of the measured response?
- What does the bass reproduced in the room sound like relative to reproduction with no room effects?
- If the user sets the subwoofer level empirically using program material, what is the objective result?
- In typical domestic rooms the speakers are acoustically far apart at mid frequencies. At lower frequencies they are acoustically close. What effect does this have on the low frequency response?
- The room response might be corrected by equalization. What is a good method for performing the equalization, and what are the perceived results?
- Only the steady state magnitude response effects have been addressed. What effect does the room have on the perceived transient response of the subwoofer?

To answer these questions three different experimental investigations were undertaken:

1. Measurement of the spatial distribution of low frequencies in eight different rooms (Section III)
2. Listening tests for the audibility of linear distortions caused by room transfer functions, and the effect of equalization on the response (Section IV)
3. Listening tests to investigate how listeners set subwoofer levels (Section V)

II. LOW FREQUENCY ROOM ACOUSTICS

Room acoustics essentially refers to the effect of reflections of sound from the room boundaries, and from large objects located within the room. Detailed discussions of room acoustics are available from several sources [1], [2]. A brief and somewhat idiosyncratic exposition will be given here as a basis for the discussion that follows. Readers who are familiar with room acoustics may wish to skip this section.

When a sound source is introduced into a room, the sound radiates from the source and bounces off the room boundaries. At each reflection some energy is absorbed and the amplitude of the wave is decreased. Room acoustics at low frequencies are best represented as modal behavior. In rectangular rooms at certain frequencies the reflection from one wall will reinforce the reflection from the opposite wall. The reinforcement causes the amplitude at that frequency to grow if the source is sustained and energy is stored such that the wave will continue for some time after the source is removed. This is a resonant behavior – the response that is obtained is identical to that of a second order resonator. Similar modal behavior also occurs in non-rectangular rooms but it is more difficult to predict what the frequencies of the modes are. Modes also occur for frequencies at which reflections from four walls, or from all six walls, reinforce each other. These modes are called axial, tangential, and oblique respectively.

The room boundaries are not perfect reflectors – energy is absorbed at each reflection. This ensures that the amplitude of the mode will not grow without bound and that the sound will eventually decay to zero after the source is removed. Since tangential modes encounter four reflections for each circuit and oblique modes encounter six reflections, they are usually less energetic than axial modes. They decay more quickly.

The frequencies of modes in an ideal rectangular room are predicted by the well-known Rayleigh equation:

$$f = \frac{c}{2} \sqrt{\left(\frac{m}{L}\right)^2 + \left(\frac{n}{W}\right)^2 + \left(\frac{o}{H}\right)^2}$$

where c is the speed of sound, L , W , and H are the length, width and height, and m , n , and o are all the positive integers, including zero.

There are millions of modes in a typical small room. The lowest frequency mode occurs for $m=1$, $n=0$, $o=0$, since by convention the length of the room is its largest dimension. In typical small rooms (volumes of 50 to 100 m³) the first few modes will be separated by about 10 Hz or so. The amount of absorption for typical walls is such that the Q of the associated resonance is high, typically about 10 to 20. This means that each resonance can be distinguished from the others. As the frequency is increased, the spacing between the modes is less and less until there are many modes with center frequencies located within the bandwidth of a single mode. The frequency at which this transition in behavior occurs is called the Schroeder frequency. Above that frequency the acoustical behavior of the room is best described by statistical parameters.

That is not the case for the range of frequencies in which subwoofers operate in small rooms. For that range the behavior is best described by analysis of the modes. It is difficult in practice to predict precisely at what frequencies and with what amplitudes the modes will affect the response at the listener's location. The exact amount of absorption of the walls is usually not known, and large furnishings and irregular room dimensions change the frequencies of the modes. It is useful to try to predict the room's acoustical performance when designing a new room, but most multichannel sound reproduction installations are put into existing spaces and this investigation is confined to those cases.

There is a significant difference in the way in which the room acoustics are perceived in addition to the need for different mathematical means to describe the room. For those frequencies that lie between room modes there is little reinforcement from reflections, so a transient signal from the source decays quickly. For frequencies which coincide

with modes there is significant reinforcement and transient signals from the source may persist at the listening position for a substantial time after the source has stopped radiating.

Interference can occur between modes. Audio signals often have a bandwidth that is large enough to allow the source to stimulate two or more modes simultaneously, and the modes have a bandwidth that, although very narrow, is still wide enough for significant overlap to occur even with sine waves. Since transmission through each mode from the source to the receiver has a different phase associated with it, it is possible for audio signals transmitted through those two paths to combine destructively and cancel at the listening position.

Below the frequency of the lowest mode, the wavelength of the sound is larger than the largest dimension of the room. At these frequencies a closed-box woofer in a room operates in a different mode than in free space, or its normal mode of operation in a room. Instead of thinking about the woofer as radiating sound, it is useful to think of the motion of the woofer cone moving out as pressurizing the room. We will call the range of frequencies in which the woofer operates in this fashion the “pressure range”. Direct radiator loudspeakers are designed to be sources of constant volume acceleration. This causes the excursion of the woofer to increase by a factor of four for each octave lower in frequency. If this mode of operation continues down to very low frequencies, the increasing excursion of the woofer would cause the response in the room to rise at 12 dB/oct towards lower frequencies. This effect has some low frequency bound since real-world physical constraints require that the excursion of the loudspeaker diaphragm be limited.

A typical measurement of the single sine wave response for a subwoofer measured in a room is:

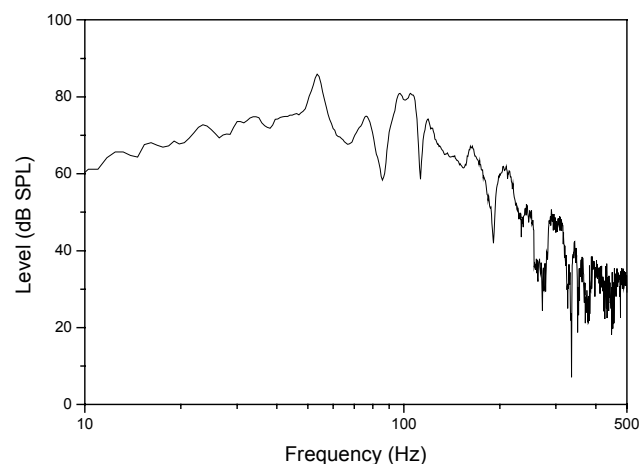


Figure 1. Typical Room Response, Including the Effect of both the Room and the Subwoofer

This figure shows the response of the subwoofer and the room. The response can be seen to roll off at low frequencies and at high frequencies, and to have fluctuations of almost 30 dB within the pass-band of the subwoofer.

The phase of the room response, as represented by group delay calculated from the directly measured phase response, is a good indicator of how waveforms will be distorted if different frequency components of the wave are delayed by different amounts when transmitted through the room. Here is a typical group delay measurement, not from the same room as in the figure above:

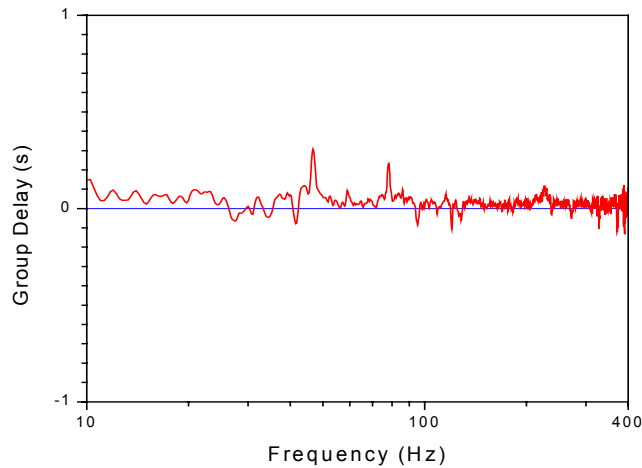


Figure 2. Group Delay of a Typical Room Response

The group delay is mostly positive, with substantial positive peaks of about 300 ms occurring at 47 Hz and 83 Hz. This indicates that signals with spectral components at those frequencies would have those components delayed by about $\frac{3}{10}$ of a second relative to the rest of the waveform.

Room Transfer Function

Direct measurement of the response at a particular position in a room gives a result that is a combination of the response of the room and the response of the loudspeaker. Naturally the result will vary depending upon the response of the particular subwoofer employed to make the measurements. In order to examine only the effects of the room acoustics, the transfer function from the subwoofer position to the reference listening position is calculated. This is done by dividing the response at the reference listening position by the free-field response of the test subwoofer.

The apparatus used to perform the frequency response measurements was comprised of the Audio Precision System Two [3], using a self-powered Velodyne HGS-12 woofer [4] as the acoustic source, and a Brüel & Kjær microphone system using the 4134 $\frac{1}{2}$ " capsule, 2619 preamplifier, and 2804 power supply. The Audio Precision measurement system performs the frequency response measurement using a Maximum Length Sequence (MLS) of order 15 (length $2^{15}-1 = 32767$) to acquire the impulse response. The impulse response is calculated from the acquired data using the Fast Hadamard transform [5], and the magnitude and phase are calculated from the impulse response using the FFT. The analyzer was operated at a sample rate of 24 kHz, allowing a measurement bandwidth of 12 kHz and a frequency resolution of 1.365 Hz, but for the purposes of this experiment only the data from 0 Hz to 500 Hz was retained. The upper end of the response range was determined by the need to compare the mid-frequency portion of the transfer function with the low-frequency portion, and by the upper limit for which the woofer could be considered to be at least approximately omnidirectional.

The AP System 2 architecture and software are such that the displayed impulse response and displayed transfer function are the data that is saved. Ordinarily the data obtained are a sub-sampled version of the actual acquired data. In order to save the actual data, the number of points for magnitude and phase must equal the displayed bandwidth divided by the resolution. No time windowing was applied, in order to acquire a long-term frequency response, and to extend the bandwidth down to DC. The MLS method produced results that were identical to those obtained by single sine wave measurement with the same frequency resolution.

The Velodyne HGS-12 subwoofer was chosen as the acoustic source since it meets the objective requirements set out in [6], within its power bandwidth, except that modulation noise is a few dB above the threshold of hearing for signals near full output. It is at least marginally portable. This subwoofer has built-in low-pass, high-pass (subsonic

filter), and polarity inversion, as well as a continuously variable level control. The low-pass filter was defeated and the high-pass filter set to 20 Hz. The polarity was set to non-inverting, and the level control was fixed to exactly its midpoint. The subwoofer was used in this configuration for all of the measurements and all of the listening experiments in this report.

In order to measure the acoustic transfer function of the room, it is necessary to know the volume velocity response of the source. It has been noted that the acoustic impedance of the room may have a non-negligible effect on the volume velocity output of the acoustic source [7]. With the source that was chosen for these experiments, the excursion is controlled by feedback from an accelerometer on the cone, in addition to being controlled by the mass of the driver. Both the accelerometer feedback and the high mass of the cone minimize the effect of a non-uniform radiation impedance on the motion of the cone.

The output of the test subwoofer was characterized by making several near-field measurements in a large acoustic space. The output was measured 1cm away from the cone with the subwoofer mounted on a stand such that the enclosure was at least 1 m from room boundaries. The resultant transfer function measurement was corrected [8] to the standard 1m reference distance for comparison to other measurements. Several such measurements were averaged to obtain the following transfer function:

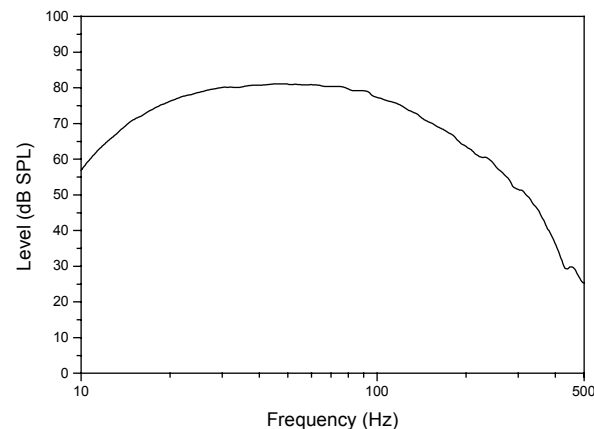


Figure 3. Transfer Function of Test Subwoofer

The subwoofer transfer function is smooth, nominally flat from about 20 Hz to 120 Hz, and rolls off uniformly outside the pass-band. The only notable aberration is a small bump in the response at about 430 Hz. The output of the test subwoofer was also characterized by multiple measurements of its impulse response and they were consistent with the frequency response shown above.

This is the subwoofer magnitude response that was used to calculate the room transfer functions that are reported in this paper.

Room Layout

The location of the full-range loudspeakers and subwoofer in the room will clearly affect the frequency response. The addition of three more speakers relative to a 2-channel stereo reproduction system applies additional constraints on the location of the loudspeakers with respect to room boundaries. Many loudspeaker layouts for a multichannel system are possible, but there are some international standards that give guidelines.

ITU Recommendation BS 775 [9] contains guidelines for the location of the five loudspeakers in a multichannel reproduction system with respect to their placement. BS 1116 [10] contains additional recommendations with respect to distance to room boundaries and to the acoustics of the room. It can be instructive to do a quick calculation regarding the recommended layout and what that implies about the position of the listener in the room.

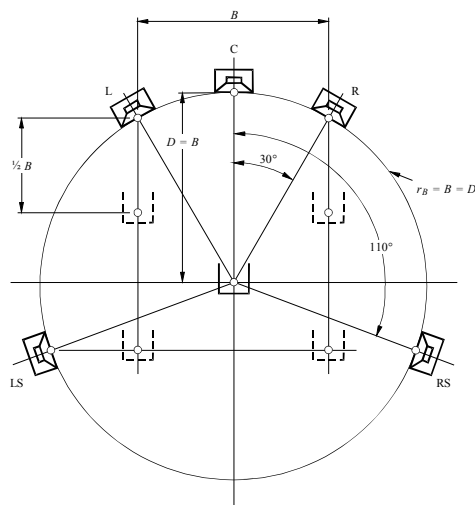


Figure 4. Recommended Speaker layout from BS 775

BS 775 recommends a frontally located center channel speaker and left and right channel speakers located at $\pm 30^\circ$ to the left and right of the center line from the listener to the center channel speaker. The left and right surround speakers are to be located at from $\pm 110^\circ$ to $\pm 120^\circ$ with respect to the centerline. If the listener is located at a distance of 2.0 meters from the loudspeakers then the width of the frontal array is 2.00 meters and the distance between the surround speakers is 3.46 m. The length of the whole array is 3.00 m. If the BS 1116 recommendation that all speakers are kept at least 1 m from room boundaries is followed, then the room must be at least 5.5 m wide and 5.0 m long. This exact arrangement would position the listener 3.00 m from the front wall, 50 cm from the geometrical center of the room.

Observation of many multichannel installations shows that they are nearly always set up with the opposite orientation; the median line from the listener to the center channel speaker is oriented along the longest axis of the room. In these cases the extra length is almost always added on to the area behind the surround loudspeakers. If another meter is added to the length of the room in that fashion then the listener is now located at the exact center of the room. This is representative of what we found in most of the rooms that were investigated. Layouts for the eight rooms that were evaluated are illustrated in section III.

The net result of these constraints is that the listener in a multichannel reproduction system is at a position that is maximally exposed to the effects of the axial room modes. The odd order axial modes have their minima at the center of the room and even order axial modes have their maxima at the center of the room. The multichannel speaker layout tends to force the listener to sit where she or he sees the greatest effect of the room modes. This apparently trivial observation has a large effect on the low frequency performance of the multichannel audio reproduction system.

Subwoofer Placement

The location of the subwoofer with respect to the room boundaries will strongly affect how it couples into the room modes. Various sorts of recommendations exist for placement of loudspeakers to achieve optimum response flatness at the listening position and there are several software programs [11] that allow the prediction or optimization of the response at the listening position. The concept of locating the loudspeakers to optimize the low frequency response was suggested by Groh [12]. This idea is very attractive since it gives the possibility of improving the subjective performance of the system without the need for purchasing any additional equipment. The effect of doing this will be investigated further in Section III.

Significant restrictions on subwoofer placement were encountered in the rooms that were investigated. Subwoofers are generally large devices and always heavy. This usually restricts location to placement on the floor and that is the only placement that was investigated.

Rooms for reproduction of multichannel sound also have additional placement restrictions. There are at least five loudspeakers in addition to the subwoofer, other audio equipment, and usually a video display in addition to furniture. Pathways to the listening position and to the equipment need to be kept clear. It is desirable to keep the subwoofer approximately the same distance from the listener as the main loudspeakers [13] in order to eliminate audible effects due to the difference in time of arrival. This usually forces the location of the subwoofer toward the periphery of the room, and in fact that was the situation in all of the rooms that were investigated.

III. SPATIAL DISTRIBUTION OF SOUND AT LOW FREQUENCIES

As seen in the previous section, the sound field at the listener's position depends upon the position of the source (the subwoofer) and the receiver (the listener). We know that, in addition to varying with the frequency, the level will also vary with the listener's position. If the low frequency response of the audio reproduction system is going to be corrected, by equalization or by any other means, it is desirable to know how correcting the response at one position affects the response at other nearby positions. What range of variation can be expected throughout the volume of space in which we might expect the listener to be located, and at what frequencies can equalization be applied?

To investigate the spatial distribution of low frequency sounds in the room, the frequency response was measured at numerous positions in a rectangular volume centered around the designated listening position. These measurements were performed in eight different listening rooms. Some of these rooms were professionally designed listening rooms or production facilities, and others were listening rooms in private homes. As a general rule the test subwoofer was located at the position already in use in the system installed in the room, or if no subwoofer was present a suitable location was chosen within the constraints of the loudspeakers, equipment, and furniture already present in the room.

A template was created to allow the microphone to be positioned at 10 cm. intervals on a rectangular 8X11 grid, and the measurements were repeated at microphone heights of 1.0, 1.1, and 1.2 meters. The 88-point grid, repeated at three heights, gives a total of 264 frequency response measurements. The size of the template was somewhat arbitrary but it was chosen to represent the reasonable extremes of where a listener's head might be if a small sofa were to be centered at the reference listening position. Each frequency response measurement required approximately 2 seconds to complete, and with relocating the microphone and adjusting it to the three different elevations, the entire set of measurements could be performed in about 45 minutes (but usually longer!).

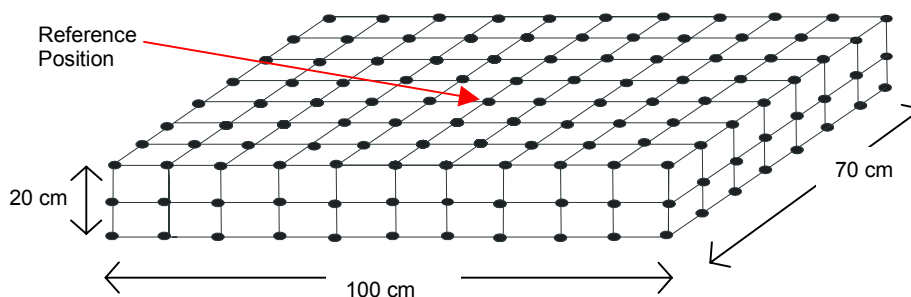


Figure 5. Array for Frequency Response Measurements in Listening Rooms

The position at the center of the volume, at height = 1.1 m, was chosen to be the Reference Listening Position, with the elevations of 1.0 m and 1.2 m chosen to represent the ear height of taller or shorter listeners.

Figures A-H show results from the spatial distribution measurements for each of the eight rooms. Each figure has three sections. The upper graph in each plot is the transfer function from the subwoofer position to the reference listening position. The second group of curves (gray) represents the difference between the transfer function at the reference position and the rest of the 264 positions in the adjacent volume. It is important to note that each group of curves is the same as what would have been measured if the woofer output were corrected by equalization to measure flat at the reference position. Finally, the standard deviation for all of the positions relative to the reference position is calculated as a measure of the variation throughout the various measurement positions.

The results of the frequency response measurements in all of the listening rooms are shown in the figures below:

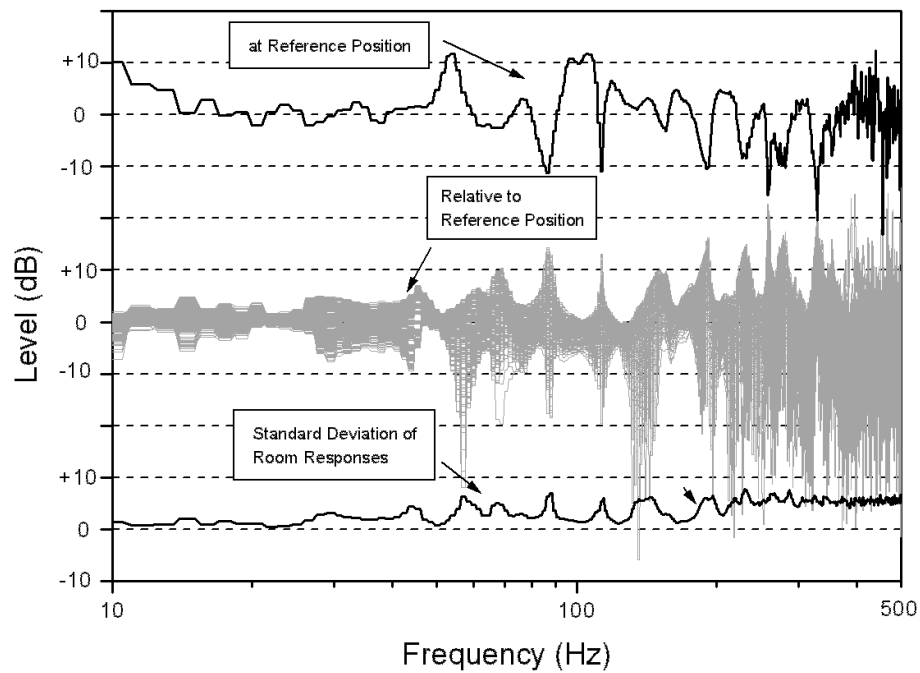


Figure 6. Listening Room A

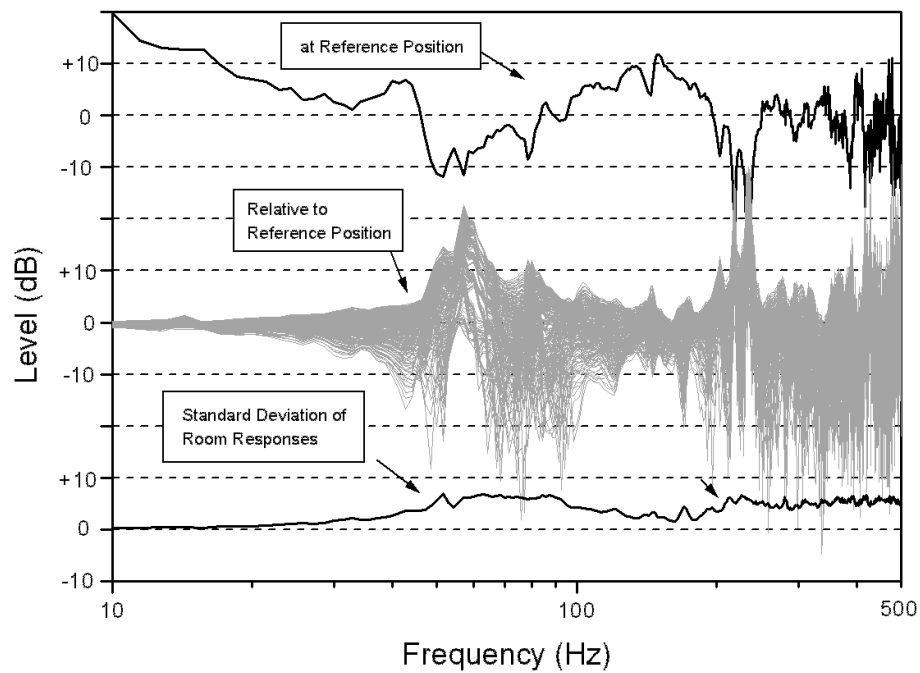


Figure 7. Listening Room B

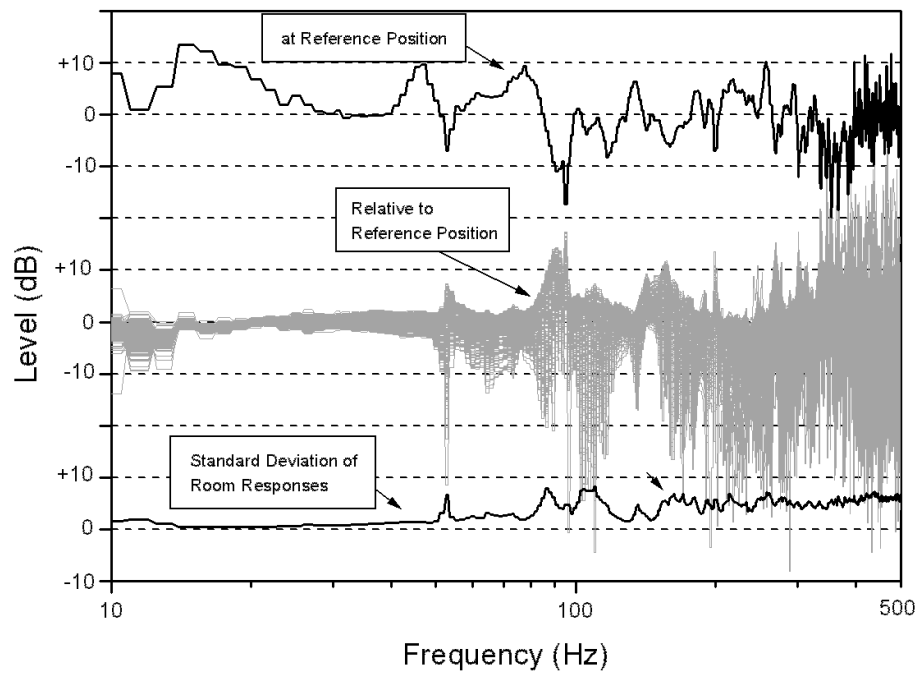


Figure 8. Listening Room C

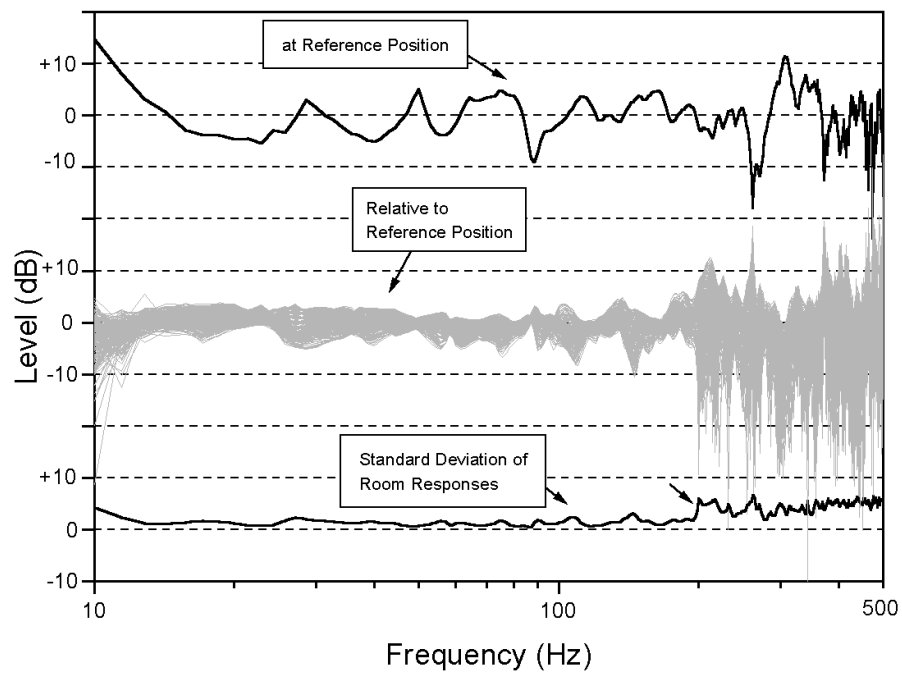


Figure 9. Listening Room D

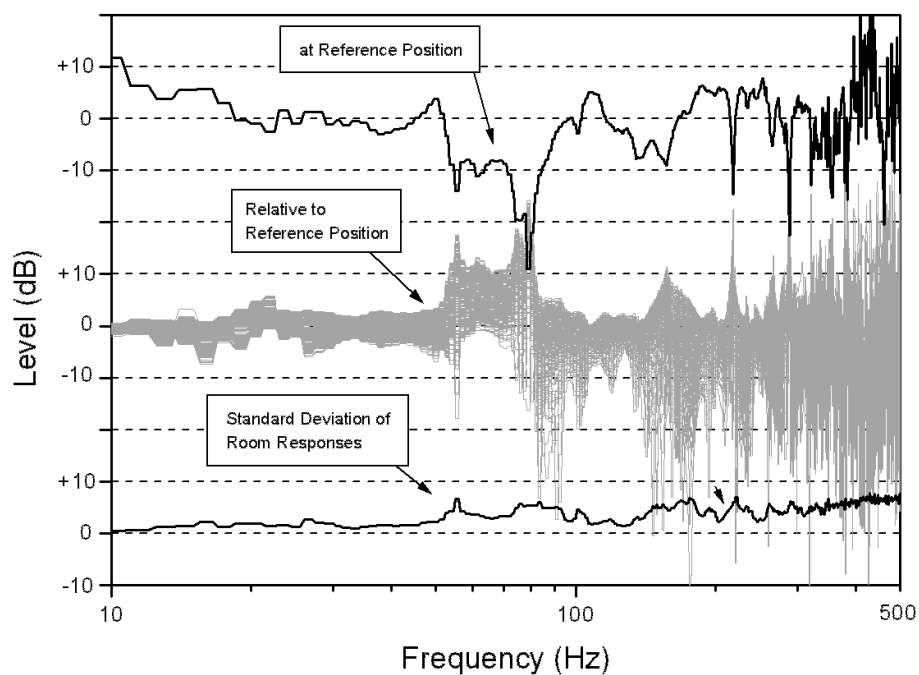


Figure 10. Listening Room E

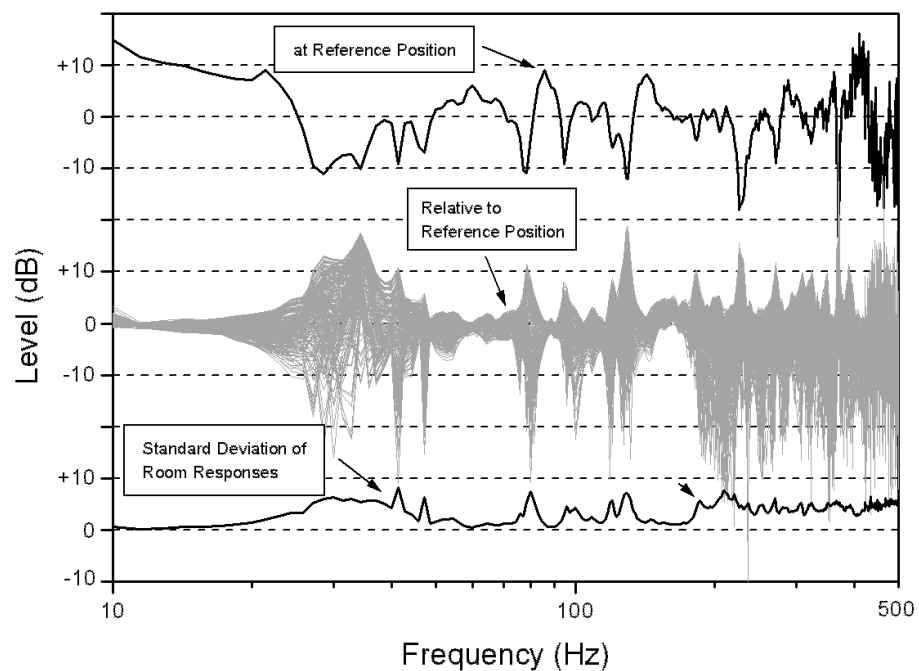


Figure 11. Listening Room F

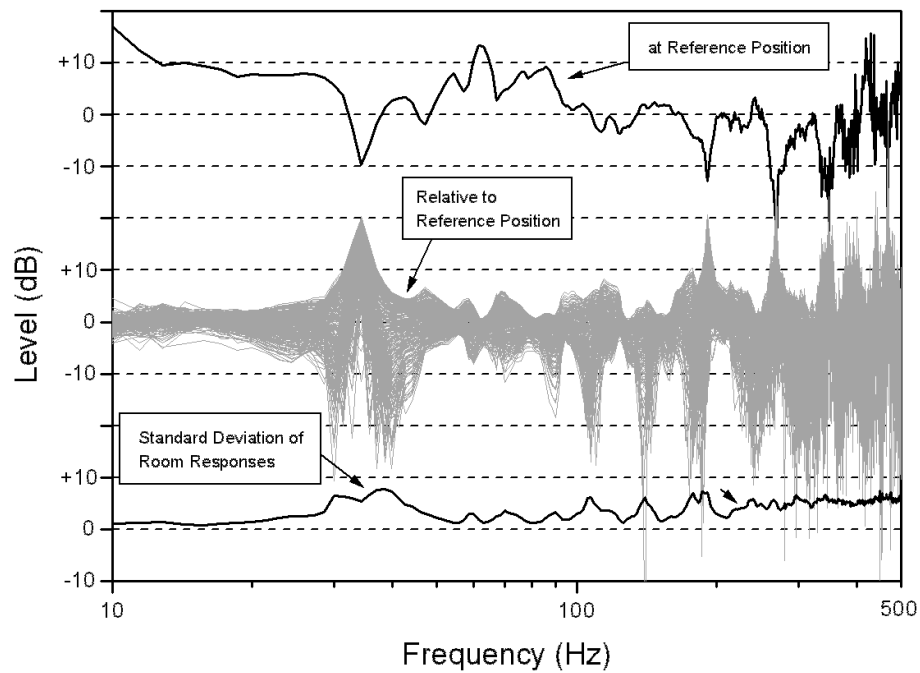


Figure 12. Listening Room G

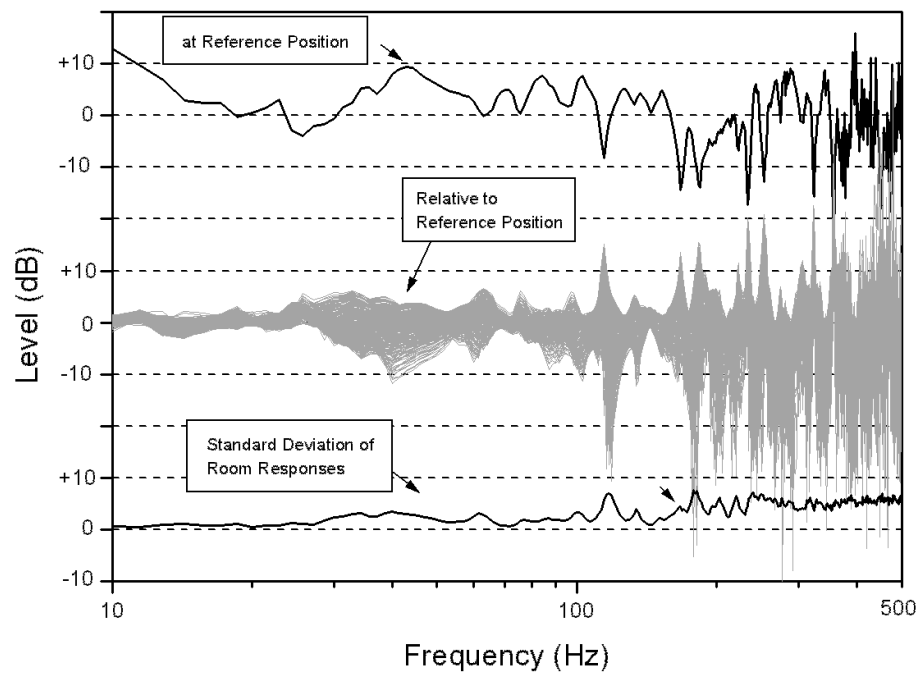


Figure 13. Listening Room H

Note: The size of the data sets generated by these measurements was initially beyond the capabilities of the various software tools used by the authors. Consequently the appearance of the plots is different for some rooms, but the frequency resolution is identical in all cases.

The transfer function measured at the reference position is the same as would have been measured if the subwoofer had had a perfectly flat volume acceleration response. The room response for any kind of subwoofer placed at the same source position can thus be calculated by knowing the response of the subwoofer.

All of the transfer functions that were measured have certain characteristics in common. The range between about 30 Hz and 200 Hz is characterized by numerous discrete peaks and dips. Above 200 Hz the peaks are too close to each other to be easily distinguished. Below about 30 Hz the response generally rises toward the lower frequencies. The family of response curves generally has more variation at high frequencies than at low frequencies. The standard deviation is used as a measure of the total variation. Its behavior as a function of frequency is particularly interesting at about 140 Hz to 210 Hz where the average variation jumps to about 6 dB. This frequency is a reasonable estimation of what is meant by the Schroeder frequency and is marked by a small arrow in each of the plots.

Listening room A (professional listening room) has high Q peaks at 53 Hz, 97 Hz, and 103 Hz. There are two marked dips at about 87 Hz and 112 Hz, which correspond with peaks in the standard deviation. The 53 Hz peak also corresponds to the rising slope of a peak in the standard deviation curve.

Listening room B (professional listening room), the smallest room measured, has a significant depression from about 45 Hz up to 90 Hz. The depression corresponds with high standard deviation so equalizing this depression is not a good idea. This room has the highest variance of any of the rooms that were evaluated. The composite set of curves for all of the transfer functions shows that the general trend is upwards at about 55 Hz. This indicates that the user would have a better low frequency response anywhere away from the reference listening position.

Listening room C (domestic living room) has high Q peaks at 45 Hz and 80 Hz. The lower peak is at a frequency with low variance, and could be expected to be equalized, but the peak at 80 Hz is at a frequency with high variance, and perhaps shouldn't be equalized. There are sharp dips at 53 Hz and 86 Hz with corresponding high variance. The low frequency rise in the transfer function stops abruptly at about 15 Hz. This may be due to the large openings to adjacent rooms. The increase in variance below 15 Hz is associated with high levels of traffic noise from outside the residence.

Listening Room D (professional listening room) is notable, relative to those measured in other rooms, in that its variance in the transitional range is more uniform than that of any other room. There is a distinct peak at 49 Hz, which is consistent throughout the measurement zone, and smaller peaks at 28 Hz, and 120 Hz. The low frequencies rise below about 20 Hz. This room is the best example of the change in the statistics of the spatial distribution, which are associated with the Schroeder frequency. That change can clearly be seen in both the envelope of the transfer functions, and in the standard deviation plot.

Room E (domestic home theatre room) has significant peaks in its response at 50 Hz and at 123 Hz. The band between 55 Hz and 95 Hz is depressed relative to the rest of the frequency range. The variance is high in the range from 55 Hz to 95 Hz. The envelope of the group of transfer functions shows that all of the error is positive. In other words, the response would be less depressed anywhere else in the volume surrounding the reference position. This room has dimensions that are quite similar to those of room A, but its acoustic measurements are very different, presumably due to the presence of the large window and generally less rigid domestic construction.

Listening room F (professional listening room) has a generally depressed region with high variance from about 25 Hz to 50 Hz. There is a substantial peak at 88 Hz, with sharp notches at 78 Hz and 95 Hz. The peak has low variance, and the notches have high variance.

Listening room G (professional listening room) has a depressed region in the transfer function between 30 Hz and 50 Hz, with associated high variance, and all of the error is positive with respect to the reference position. There is a very substantial high Q peak at 52 Hz that corresponds with a minimum in the variance.

Listening room H (domestic living room) has a relatively smooth response with respect to the other rooms. There is a broad bump centered on 40 Hz, with a moderate increase in variance at the frequencies associated with the bump.

In the range from 30 Hz to 150 Hz, there are usually one or two peaks that are higher than the rest. These peaks are the primary candidates for equalization. Comparison of the measured transfer functions to modal analysis for the room indicates that the second order axial mode due to length is usually the dominant acoustical influence at low frequencies. Occasionally there is a trough in the response and all of the variance is positive with respect to the reference listening position, indicating that the overall performance would be improved if the reference listening position would be moved.

We assume that if equalization is to be performed that it would be best to do so only at frequencies where the response does not vary significantly through the volume immediately adjacent to the reference listening position. There are several cases where this rule implied that equalization should not be applied to one of the prominent features in the room response.

All of the rooms show a substantial low frequency rise below 30 Hz. This rise is most prominent in the smallest room (B). The effect stops abruptly in room (BDG) at about 14 Hz, presumably due to the present of several large openings to adjacent rooms. The phenomenon of the low frequency rise has been observed previously in small rooms [6], [23], and in automobiles below 70 Hz [24]. Several of the professional listening rooms were measured both with the door closed, and with it open, and in for all rooms the extreme low frequencies are increased by about 1 to 3 dB with the door open.

The resonances in domestic rooms usually show a lower Q than those in rooms designed for professional use. This is presumably due to the generally less rigid construction and the presence of windows in domestic rooms. Analysis of the various groups of transfer functions and standard deviation calculations gives the following general conclusions. Throughout the test space there are sometimes frequency aberrations that lend themselves to equalization. These are usually peaks in the transfer function with an associated low standard deviation. Areas of higher standard deviations are probably not good candidates for equalization, because the response at positions adjacent to the reference position will not be improved to the same degree as the reference position, and may sometimes be degraded. In areas of high standard deviation, the best approach may be to equalize for a general trend within the volume, or not to equalize at all.

It has been reported [14] that peaks in the response are less susceptible to spatial variation than are dips in the response. While this is generally true, there are several exceptions visible in the measured data. In some cases a measured peak in the transfer function translates to a dip at another position close by.

An example of one such situation is taken from the data for room "H". The figure below shows the response at the reference listening position, with a substantial peak, and the response at a selected position nearby.

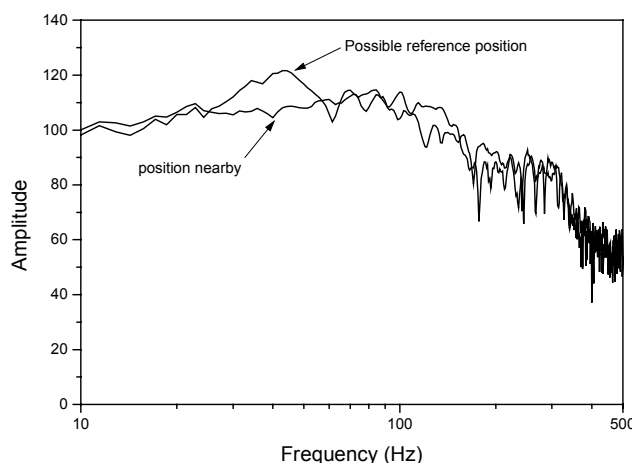


Figure 14. Example of a Bad Peak to Equalize

If the peak at the reference listening position is equalized, a dip will be created at some positions within the adjacent volume.

Placement Equalization

As was noted in section I, the placement of the subwoofer and of the listener affects the response at the listener's position. Optimally choosing the placement of both subwoofer and listener can result in substantial improvement in the transfer function by carefully controlling the degree to which the subwoofer is coupled into the room modes, and the degree to which the listener is exposed to the excitation of room modes.

This type of placement equalization was done for room "A". A new position was found for the listener and for the subwoofer that significantly improves the response. This was accomplished by moving the subwoofer to a position slightly farther out of the corner and with different distances to the front and side walls, and by moving the listener very slightly forward with respect to the original reference listening position. The new positions are marked Sub 2 and Position 2 in the room outline figure for room "A". The new position for the listener is probably unsatisfactory for stereo or multichannel listening because it places the listener too close to the front speakers and too far from the surround speakers.

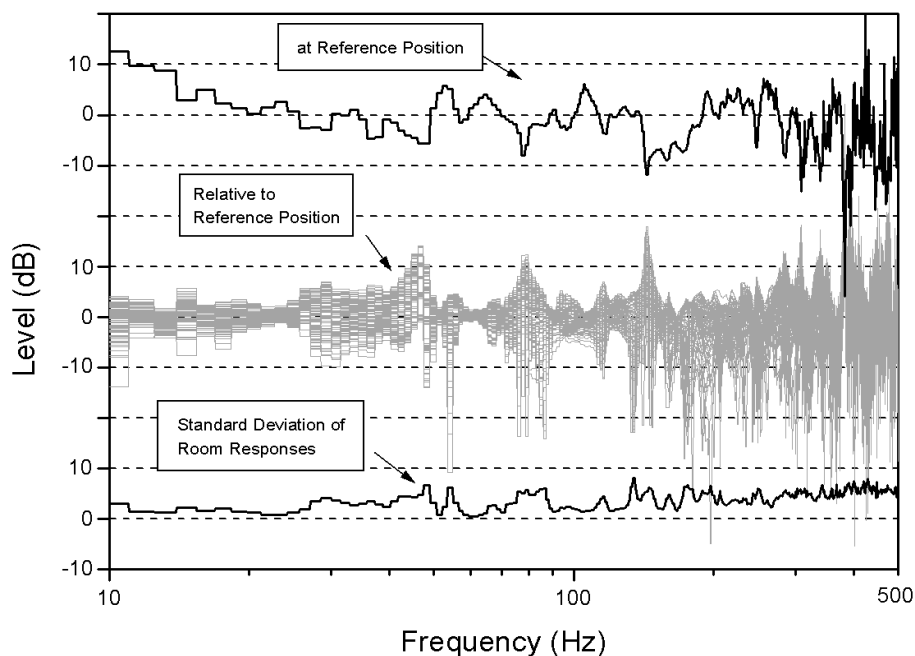


Figure 15. The Effect of Placement Equalization on the Spatial Distribution

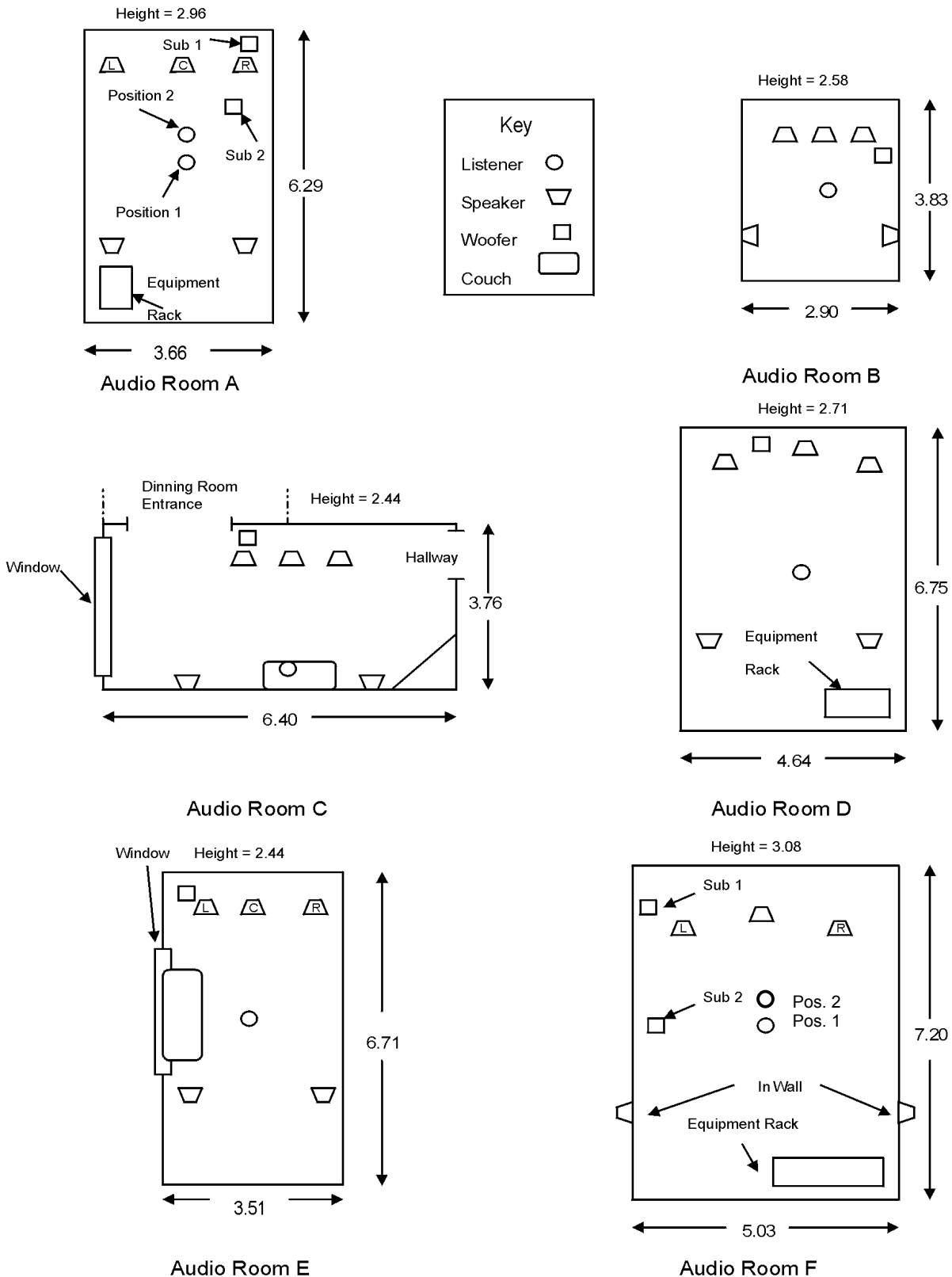
The original position has a transfer function with variations of approximately ± 12 dB (figure 6) and the optimized position has variations of only about ± 6 dB (figure 15). The new position has a significantly improved response.

The question arises: To what extent does placement equalization of the response affect the spatial distribution?

A set of spatial measurements similar to those described in the previous subsection were made to assess what changes in the spatial distribution have been caused by the change in source and receiver position. The measurements were processed and displayed in the same fashion as in figures 6-13, and the results are displayed in figure 15. Detailed inspection of the range of the individual transfer functions, and of their standard deviations, shows that the variation is slightly improved at some frequencies and slightly worsened at other frequencies. The distribution of transfer functions has changed in that the peaks in the standard deviation plot now correspond almost exactly with the peaks in the transfer function.

While there is no improvement in the spatial distribution after placement equalization, it should be noted that since the response at the reference position is improved, and since the deviations in the response throughout the test volume are measured with respect to that improved response, then on average the response has been improved throughout the volume.

Outlines of listening rooms



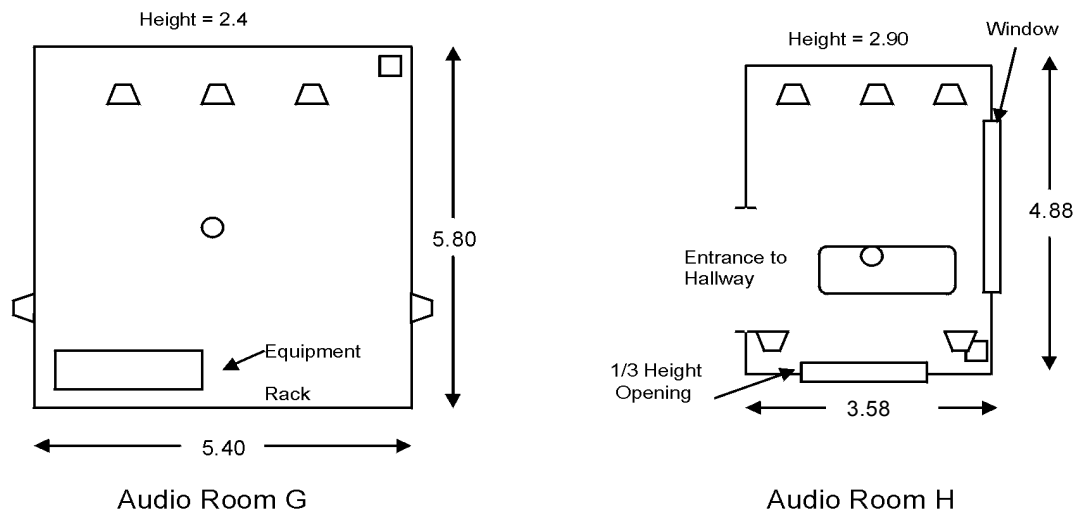


Figure 16. Sketches of Loudspeaker Placement in Eight Listening Rooms

IV. SUBJECTIVE EFFECT OF LOW FREQUENCY ROOM ACOUSTICS

The subjective effect of low-frequency response aberrations will be different depending on whether the program is continuous or transient, and whether the spectra of the program matches with the frequencies of the aberrations in the room response. All signals will have their spectral content altered by the room transfer function. Tonal sources will have some frequencies emphasized relative to others. Noise-like sounds can be changed from being without timbre, to having a tonal quality. It is not expected that noise-like signals will be affected by the phase component of the room, but other types of signals will have their waveforms changed by phase distortion.

It can be seen from the measurements presented in the previous section that the objective effect of the room acoustics on the objective low frequency performance is profound. While electrical measurements of audio systems typically show errors in the magnitude response of a fraction of one dB, and phase errors of less than a few degrees between 30 Hz and 120 Hz, the errors introduced by room acoustics are much larger. The typical range of the magnitude response is as much as ± 15 dB, and there are group delay differences amounting to as much as ± 400 ms [23]. What is the subjective effect of these large errors? Does equalization of the magnitude response achieve a result that is more like the original program material?

The subjective evaluation of the low frequency distortions introduced by the room acoustics was performed in two phases. In the first phase a wide variety of program material was compared and evaluated in order to spot general trends, and to locate material that showed interesting behavior for further consideration in formal listening tests. In the second phase a formal listening test was performed in order to estimate the amount of change in the quality of the low frequency reproduction due to room acoustics, and the improvement that can be accomplished by equalization.

Description of Low Frequency Sounds

Low frequency sounds vary in terms of their fundamental frequency, spectral occupancy and waveform envelope. A survey of low frequency sounds in music and theatrical soundtrack recordings was made. During this process it was found to be convenient to group low frequency sounds into the following general groups:

- Tones – organ notes, synthesizer notes, test tones
- Bursts – plucked strings, drums, thumps, test bursts
- Noise-like – explosions, rumbles, earthquakes

In terms of what is heard, rather than a physical description of room acoustics, the dense overlapping modes at mid and high frequencies produce a reverberation that can, in part, be distinguished from the direct sound. At low frequencies, below the Schroeder frequency, the “reverberation” begins only a few cycles into a note, and what is heard is a change in the character of a note. Low frequency sounds reproduced in small rooms are commonly referred to as “boomy”, or “colored”. Sometimes there is a loss of articulation.

Room Impulse Response

In the process of measuring the transfer functions of the rooms in section II, Room Impulse Responses (RIRs) were measured which represent the transfer function from the subwoofer location to the principal listening location. These impulse responses were measured using the same equipment and setup as was used to characterize the transfer function. A Maximum Length Sequence (MLS) from the Audio Precision System Two was applied to the source transducer at a level that had been tested to be within the linear range of the device. The sound was captured at the listening position using a Brüel & Kjær microphone system consisting of a 4134 ½” capsule, 2619 preamplifier, and 2804 power supply.

The microphone system inverts polarity, requiring that the resultant impulse responses be inverted after acquisition. The output was analyzed by the Audio Precision System Two. Although this system is primarily designed to measure and display transfer functions, if the number of data points is set to be equal to the number of samples in the MLS, then the complete IR is displayed and can be saved to disk. For some rooms it was necessary to perform multiple measurements and to average the resultant RIRs in order to obtain a sufficiently good signal to noise ratio.

The impulse responses that were acquired were done with a sample rate of 24 kHz. Because the analyzer used has an MLS of order 15, only 32767 samples ($2^{15} - 1$) of the whole impulse response could be acquired. This gave a time record with an approximate length of 1.365 seconds, which was found to be longer than the decay times of any of the rooms evaluated.

In order to easily perform convolutions with program sources which were originated at sample rates of 44.1 kHz and 48 kHz, the acquired impulse responses were sample rate converted up to 44.1 kHz and 48 kHz. A library of RIRs was generated which included the original RIRs for each of the rooms at the 24 kHz sample rate, and at 44.1 kHz and 48 kHz. The two RIRs that were used in the listening tests are displayed in the figure below:

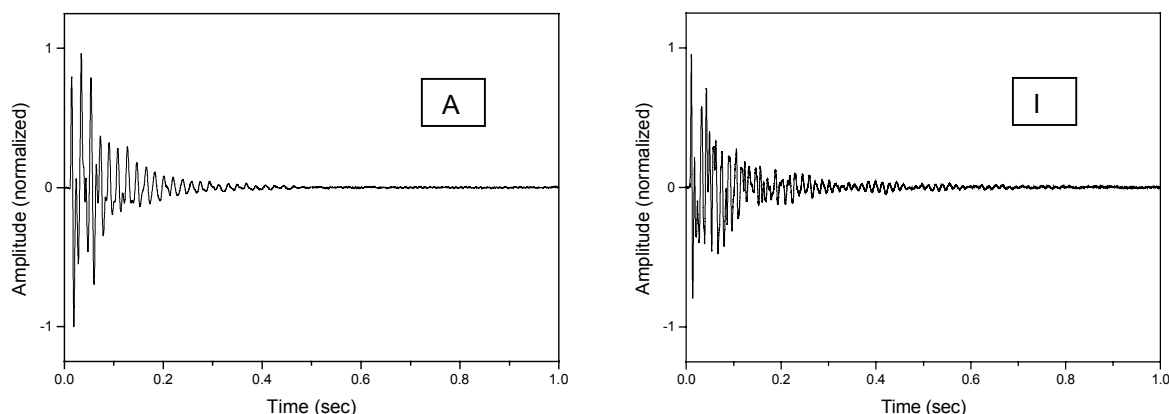


Figure 17. Room Impulse Responses for Rooms “A” and “I”

Both impulse responses ring for several tenths of a second. Listening room “A” shows a clear progression from a broad-band characteristic to an exponential decay of primarily one frequency after about the first tenth of second. The impulse from listening room “I” has a similar characteristic but the decay takes longer and at least two frequencies are present. Beating between the two modes is apparent with a period of about 150 ms. Listening room “I” is not one of the eight rooms evaluated in section II, but rather one of several other rooms which were measured less completely. Its RIR was used in these experiments because it has interesting properties.

Parametric Equalization

The effect of the room resonances at a particular listening position can, in principle, be equalized exactly. There are several reasons why this may not be a good goal. Firstly, there are narrow notches in the typical response which, if equalized exactly, would require substantially more power output from the system to correct. Secondly, as shown in the section on spatial distribution of low frequencies, correction of the response at a single position will always result in an uncorrected error at other positions within the listening window. Finally, there is the comparison with what can be done in the rest of the audio spectrum. If the low frequency response is corrected exactly, but the high frequencies aren’t corrected, or are only broadly corrected, then there will be reverberation at high frequencies, but no reverberation at low frequencies. Essentially all recordings are designed to be reproduced in a reverberant environment, because they are produced in reverberant environments. Eliminating all low-frequency reverberation could have undesirable results.

A better goal for low frequency equalization is to correct the part of the LF transfer function that causes audible artifacts by reducing the length of time that particularly energetic modes reverberate to match the reverberation time of the rest of the spectrum. The specific algorithmic approach is:

- compute the FFT of the RIR
- identify the most prominent peak
- calculate the frequency, Q (from the width of the spectral peak), and level
- apply a complimentary filter to attenuate the peak of the selected resonance to the average level of other peaks
- compute the FFT of the equalized RIR
- continue the iteration process

This algorithm has the disadvantage that estimating the Q of a resonance by examining the spectrum is an inaccurate process because the frequency resolution of the response measurement is too coarse. A specific resonance can be lower in level, but have a high Q . In that case it will have less of an impact on the spectrum of the whole impulse response, but will continue to be excited longer than other resonances.

An alternative variation of the equalization algorithm is:

- derive a spectrogram with a short analysis window (about 100 ms time resolution)
- identify the resonance that has the longest decay time
- calculate the frequency, Q (based on the decay time), and level
- apply a complimentary filter designed to reduce the Q of the selected resonance to the average Q of the group
- compute the spectrogram of the equalized RIR as in the first step
- continue the iteration process

This will result in an impulse response in which the group of resonances decays at a more uniform rate, and the length of the RIR will be decreased. The second algorithmic approach was used to equalize rooms “A” and “I”. The RIRs of the equalized rooms were used in the experiment described in the following subsection. The equalized RIRs are shown in the following figure:

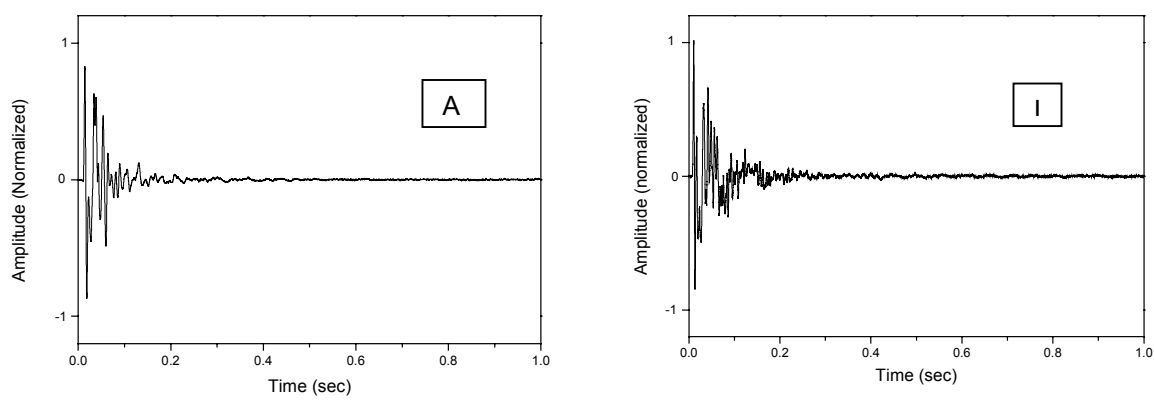


Figure 18. Room Impulse Responses for Rooms “A” and “I” after equalization.

The equalization can be seen to have significantly reduced the amount of time that the room rings after excitation.

The sound that is heard by the listener is a combination of the effects of the loudspeaker and the room acoustics applied to the program source or test signal that is being reproduced. In order to evaluate what the subjective effect of the room acoustics is, it is necessary to be able to listen to that effect in isolation. It would be desirable to be able to turn the room acoustics off, and then to listen to the room effects as they were restored in stages. This is not generally possible to do. An alternative approach is to apply the room acoustics to program material for headphone reproduction. This allows the room transfer function and an equalized version of the room transfer function to be subjectively evaluated in isolation.

The transfer function for headphone reproduction can be altered to that of the room by convolution of the program signal with the Room Impulse Response (RIR). It is important that the headphone transfer function not contribute audibly to the total transfer function, or it should be corrected to eliminate it as a source of error.

The headphones used for these experiments were Stax Kogyo type Lambda Pro. These headphones are of open-air circumaural type, with electrostatic transducers. The Stax Lambda Pro are of the same type as employed in [6], where they were demonstrated to have extremely low distortion and extended low frequency response. The response of the headphones used in these experiments was measured by placing them on a dummy head, B&K type 4128 Head And Torso Simulator (HATS). That response is shown in the figure below:

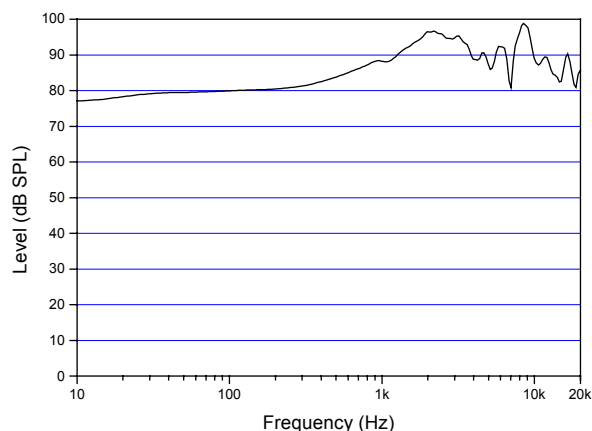


Figure 19. Magnitude of Transfer Function of Stax Lambda Pro Headphones

The magnitude response can be seen to be extremely uniform at low frequencies. The level at 10 Hz is only attenuated by about 2 dB relative to the level at 100 Hz. The substantial rise at 2.5 kHz is an expected aspect of the response that is due to the ear canal resonance in the B&K 4128 dummy head. The low frequency response achieved when the headphones are placed on a real listener can vary somewhat depending upon how good the seal is between the headphone cushions and the listener's head. If there is a leak the low frequency response will have a second order high-pass response with a small peak somewhere in the range of 50 Hz.

Experiment Design

The basic methodology of the experiment was to split the frequency band of the program material as would be done by a crossover in an audio reproduction system. After filtering, the low-pass portion of the program was convolved with the RIR for one of the rooms. Convolution was initially done with a proprietary DOS-based filtering program, but the results presented here were prepared using the Aurora Real Time Convolution plug-in for CoolEdit Pro [15]. The time delay of the program segment was compensated to correct for the convolution process and for the delay in the RIR. The low-pass convolved portion of the program was normalized such that its rms level was made to be the same as before convolution. Then the low-pass and high-pass portions of the program were summed back together.

The crossover used is an 8th order Butterworth-squared (Linkwitz-Riley) filter at 120 Hz. The high order filter was chosen in order to limit the extent to which the convolved and un-convolved portions of the spectrum overlap. The result of the filtering process is that the magnitude of the sum is flat, but that it has a 4th order all-pass characteristic. The phase distortion of such a filter may possibly introduce an audible waveform distortion. Informal listening tests were conducted in order to assess the audibility of that distortion using the program material selected for the experiment. No audible effect of the crossover was found for any of the program material. The figure below illustrates schematically the process by which the program material was prepared.

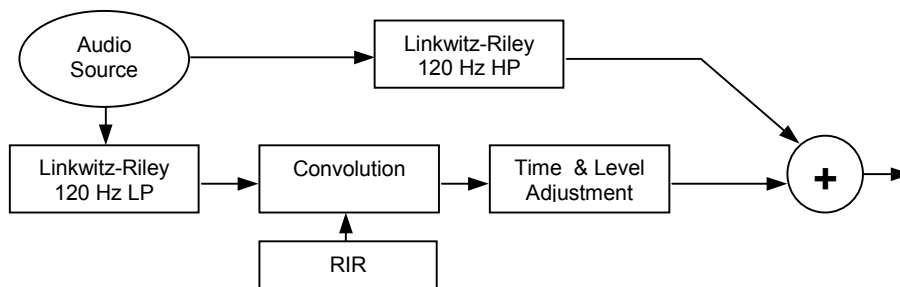


Figure 20. Block Diagram Showing Process of Construction of Listening Material

Perhaps the most significant conceptual problem with this experiment is that we study the effect of the low frequency room acoustics in isolation from the effect of room acoustics at mid and high frequencies. Our primary justification for this simplification is that we are studying the reproduction from a transducer that has an application that is fundamentally limited to a maximum frequency of 100 Hz to 200 Hz at most. If we were to have attempted to apply room reverberation to the high-pass portion of the program, the choice of an appropriate Room Impulse Response would be problematic at best.

Formal Listening Test:

A formal listening test was performed to specifically evaluate the audibility of artifacts introduced by low-frequency room acoustic effects. Program material was chosen from the larger group evaluated in the informal listening tests to specifically highlight some of the things that were noticed in those tests. The test was designed to detect whether the listeners heard the effect of the room, and whether parametric equalization could compensate for those effects, and to what degree.

Real program material has the aspect that usually several or many instruments are playing at one time. This restricts the audibility of artifacts in the reproduction, since the spectrum of one instrument may cover up a distortion in the frequency domain or time domain behavior of another instrument. It is useful to use simple signals to test the subjective performance in these cases.

Selection of Listeners

Seven test subjects were selected from individuals with extensive experience in high-fidelity sound reproduction and mixing or formal training in critical listening. All of the listeners were male and between the ages of 25 and 35. None of the listeners reported having had any known hearing pathology. Training was performed to ensure that all subjects could correctly repeat the results of a low frequency level setting exercise.

Performance in subjective listening tests is known to improve with training. It has been shown that there is a correlation between listener selection and variance of results in subjective evaluation. It has been reported [19] that the use of highly experienced listeners can reduce the number of listener required to produce a certain confidence interval by as much as a factor of seven relative to using inexperienced listeners.

Program Material:

Material with low-frequency transients was selected because the preliminary informal testing showed that room acoustic effects significantly changed transients. The specific instruments that were found to be most affected were electric bass, upright bass, drums, and synthesizer.

For this test four pieces of material were chosen. Three of these pieces are commercially available.

Program A - Diana Krall, "Love Scenes", (Tr. 11, 0:00 - 0:51 GRP Records, IMPD-233)

Program B - The Boston Bass Disc", (Tr. 10, 34.5-39.0, Windham Hill Records, WD92-18)

Program C - Kick Drum sample, Sony DRE-S777 Demo CD, (Tr. 16, 0:00.0 - 0:05.0)

Program D - Gaussian burst test signal

The fourth program is an artificial test signal. With real program material usually several or many instruments are playing at one time. This restricts the audibility of artifacts since the spectrum of one instrument may cover up a distortion in the frequency domain or time domain behavior of another instrument. It is useful to use simple signals to test the subjective performance in these cases.

A group of Gaussian-windowed bursts were prepared in order to test both the objective and subjective behavior of the low-frequency room acoustic effects. These bursts were prepared using software tools in a PC. The bursts were designed with center frequencies on $\frac{1}{12}$ octave intervals from 200 Hz down to 10 Hz at a peak level of -20 dBFS.

This spacing gives a frequency resolution of 5 Hz at 100 Hz and 0.6 Hz at 10 Hz. A 50 Hz burst, representative of all of the bursts, and its power spectrum are displayed below.

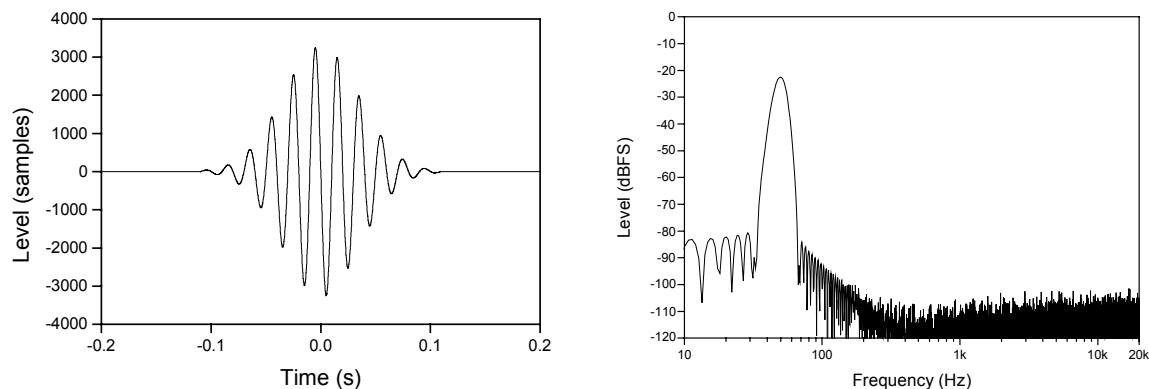


Figure 21. Waveform and Power Spectrum of 11-cycle 50 Hz Gaussian Burst

The small side lobes in the spectrum are due to the finite length of the Gaussian window and are below the threshold of hearing at the reproduction levels employed in these experiments. The test sequence presented to the listeners was prepared by concatenating individual bursts with 1 second of silence between bursts. The bursts decrease in frequency starting at the highest frequency and progressing to the lowest frequencies.

For program material A - C the process of filtering, convolving and summing the program material was used. For program D, the Gaussian-windowed bursts, it was not necessary to perform the high pass and low pass filtering stage since the material is inherently band limited to 120 Hz.

Test Setup

The test setup included a personal computer equipped with a quiet power supply and silent CPU cooler, proprietary ABX comparison software, Stax Kogyo type Lambda Pro headphones, and a Digital Audio Labs Card Deluxe professional sound card. A block diagram of the setup is shown below:

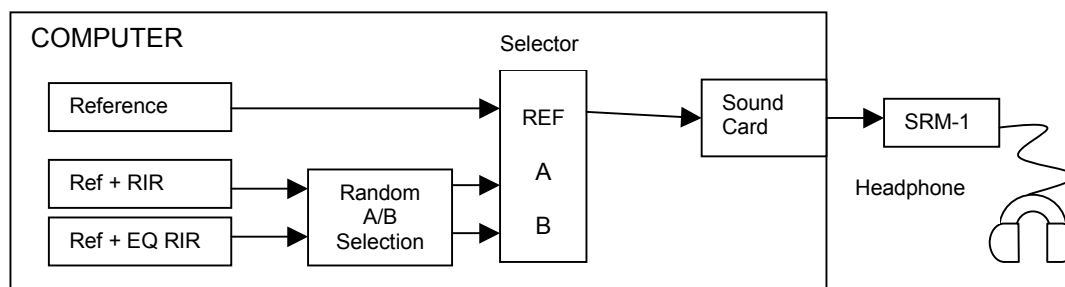


Figure 22. Block Diagram of Test Setup for Listening Tests on Bass Quality

The test subject was seated in a quiet room with a background noise level less than 30 dB(A).

Listening Test Methodology:

The subjective effect of low-frequency aberrations caused by room acoustics was evaluated with and without equalization. The basic experimental protocol was to provide the listener with three versions of same program material:

- An unaltered reference program.
- The reference program convolved with the low-frequency RIR.
- The reference program convolved with an equalized version of the same low-frequency RIR.

The listener was asked to compare the two processed audio programs to the reference program. Proprietary software was used that allows for rapid and seamless comparison between all three audio programs at the same time. The software provides the listener with three buttons; REF, A, and B, which select between the reference and programs A and B. Basic playback controls are provided that allow the listener to stop, play, loop, and select a subsection of the audio program for more critical review.

REF was always assigned the reference material while A and B were randomly assigned one of the two audio selections. The listener was allowed to compare each audio selection any number of times.

Each listener was provided with written and verbal instructions before the test. The instructions described the test setup, evaluation criteria and types of aberrations that they might hear. Additionally, training was preformed with each listener to ensure that they were familiar with the sounds in the subwoofer frequency range.

Listeners were instructed to compare the material based on the following question:

- How different are the two test programs from the Reference?

This test measures the audibility of the room transfer function aberrations relative to the reference, and whether the equalized room is audibly closer to the reference than the un-equalized room. Listener preference as to which material sounded better or worse was not considered. With some program material the sound quality with room resonances present may possibly be preferred relative to the original.

Comparison between the two programs was made using a slightly modified version of a standard 5-point subjective comparison scale. For this test a range from 5.0 (Imperceptible) to 1.0 (Very Different) was used. The results of the comparison were unscrambled and saved by the program.

Testing was preformed using the four audio selections and RIRs from two rooms, “A” and “I”, as described previously. This creates a total of eight comparisons.

Results from Informal and Formal Listening Tests on Bass Quality

The listeners were asked to rate the sound quality of the low frequency reproduction of two processed samples relative to the original unmodified program. One sample was the program convolved with the Room Impulse Response, and the second sample was also convolved with the room impulse response but also equalized with a parametric filter designed to partially correct the response aberrations added by the room, as described in the section on equalization.

The ranking scale was:

- 5 Identical to original
- 4 Bass quality slightly different than original
- 3 Bass quality moderately different than original
- 2 Bass quality significantly different from original
- 1 Bass quality very different from original

The scorings of the listeners were analyzed across the variables of listener, room, and program. The mean scorings of all of the listeners are presented for each combination of room and program in the figure below:

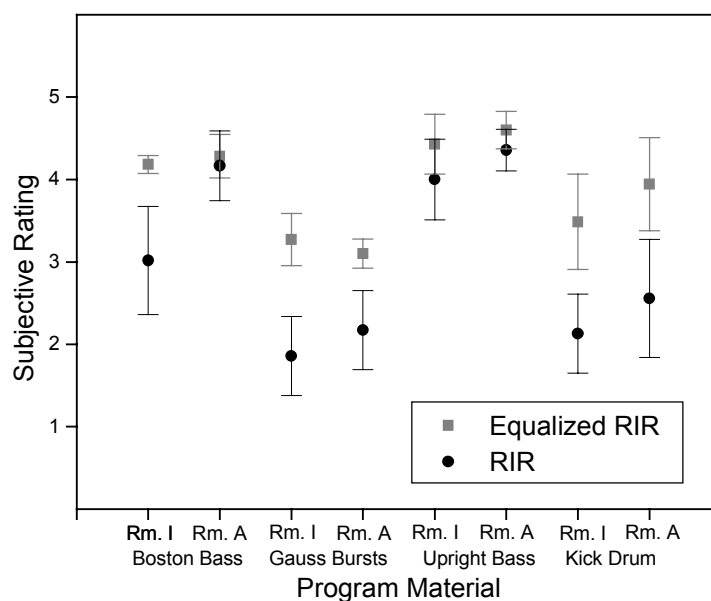


Figure 23. Mean scores for “Equalized” and “Not equalized” for Four Programs and Two Rooms

Each combination of program and room impulse response was different from the original. In general, listening room “I” produced a larger change than did “A”, but not with statistical significance, and not for every type of program. Equalizing the room transfer function produced a result that was more like the original in five of the eight combinations of rooms and programs. Analysis of the ordering of the scoring in the raw data for the each listener indicated a stronger difference between “non-equalized” and “equalized” than can be seen from the mean scores. Both the non-equalized and equalized presentations were always heard to be different than the original, and in seven out of eight combinations the equalized version was perceived to be more like the original than the non-equalized version.

The typical kinds of distortions that were heard during both the informal and formal phases of the listening tests fall into the following general classes:

- Some frequencies were emphasized or de-emphasized relative to the original source – some notes in a series were either too loud or lost; some frequencies in a band were obviously louder.
- Modulation effects and echoes – modulation was heard on the decay of notes, or notes that were of short duration were heard to be repeated one or more times.
- Pitch change – notes that were of short duration were heard as having a different pitch than the original.

Of these three observations, the first is obviously expected, and the second two observations seem less likely.

The explanation for the echoes is as follows: In many situations the notes in the program are relatively broad-band. They can be modeled as sine waves with an envelope that limits the observed duration of the note to 10 to 20 cycles. This modulation causes the spectrum to widen. The energy of the note is carried by several adjacent modes in the room. The modes each have a second-order, high Q transfer function with associated magnitude and phase transfer functions. The amplitude and phase of the components carried by each mode will in general be different, and they can interfere at the receiving position (the listener). The Q of low frequency room modes is high enough that a 10 cycle note can be stretched to 30 or more cycles, and then the interference can cyclically reinforce and attenuate the single note such that 2 or 3 distinct bursts appear at the receiving position. As an example; 30 cycles at 40 Hz last for 750 milliseconds. Interference can cause most of the middle 10 cycles to cancel resulting in two 10-cycle bursts spaced apart by 250 ms.

The Gaussian bursts described earlier in the subsection on Program Material proved to be good for illustrating both the time-domain echoes and pitch-changing characteristics heard in the rooms. An example of the audible “echoes” for a single Gaussian burst at 67.27 Hz convolved with the room impulse response of Listening Room “I” is shown in the figure below.

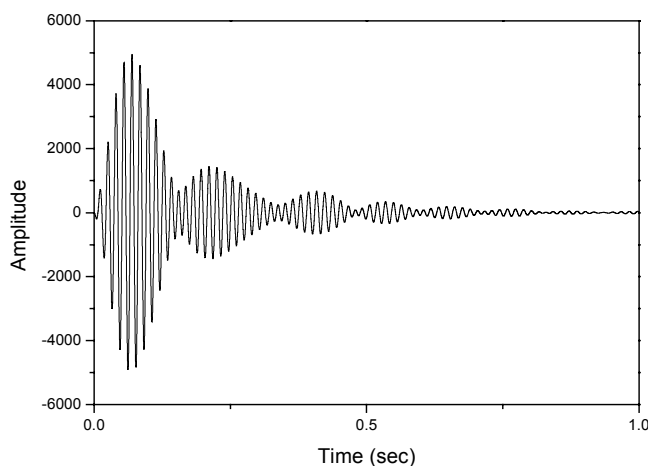


Figure 24. Amplitude of Output from Listening Room “I” with 67 Hz Gaussian Burst

Several bursts of similar shape, but slightly lower in amplitude, are visible in the output. Room “I” has peaks in its room transfer function at 62.5 and 71.3 Hz. This burst falls between those two frequencies. Since peaks in the response are generally sparsely distributed at low frequencies, there are always sounds that will fall in between those peaks. The behavior shown in the figure is similar to what is seen at many frequencies in all of the rooms that were evaluated.

The phenomenon of beats in the decay of a sine wave in a small room has been known for at least 50 years. Knudsen [16] described beats in the decay of a 96.7 Hz sine wave reproduced in a room with modes at 92.8 Hz and 99.8 Hz. The resultant beats were separated by about 1/6 s. Salava [7] describes several experiments that show modulation effects at the ends of rectangular bursts. Linkwitz [17] describes various modulation effects at the end of groups of four low frequency bursts reproduced through in room.

The apparent pitch change is explained by the selectivity of the filters in the room response. Typical short notes in music have, despite the perceived discrete pitch of the note, a spectrum that spans about one third of an octave. An example is shown in the figure below, of a single note from an upright bass solo by Christian McBride (Diana Krall, Love Scenes, “My Love is”).

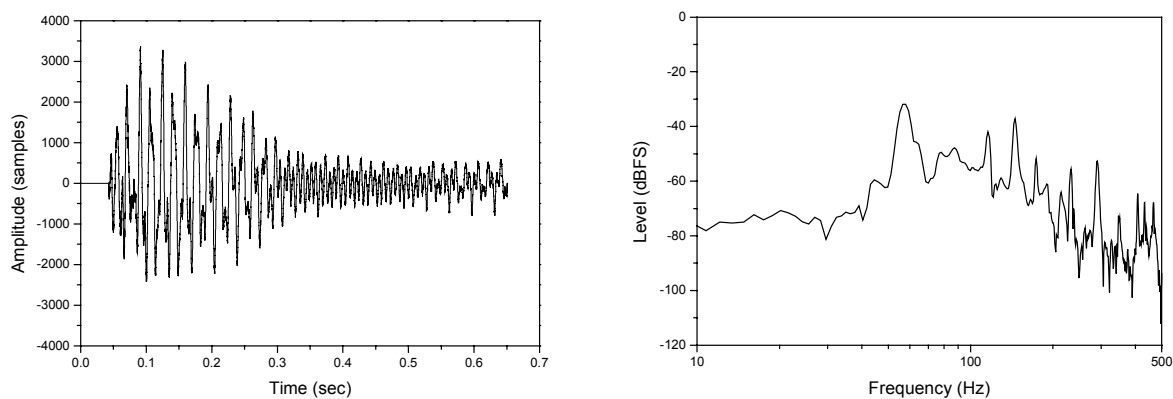


Figure 25. Waveform and Spectrum of Note A1 of Plucked String Bass

The rapid attack and short exponential decay of the note broaden its spectrum. The width of the fundamental of the note can be seen to be roughly similar to the spectrum of the Gaussian burst in figure 21.

If this relatively broader spectrum is applied to a very high Q filter, the filter will select that portion of the spectrum that is within its band-pass. If the perceived pitch is at the center of the spectrum of the signal (musical note), when the spectrum is changed by passing through the filter system of the room transfer function, then the perceived pitch may also change.

Figure 26 shows the frequency spectrum of six successively lower frequency Gaussian-windowed bursts, before and after convolution with the RIR of listening room A. The transfer function of listening room A has a high Q peak at 53.8 Hz. This peak causes significant changes in the spectrum of the convolved bursts. Each of the six individual plots in Figure 26 shows an original burst (smooth spectrum) and the convolved result (distorted spectrum). The plots are order from left to right with burst frequencies of 67.1 Hz, 61.9 Hz, 59.2 Hz, 53.8 Hz, 48.0 Hz and 44.8 Hz. The center frequency of each burst is shown before and after convolution. A centerline at 53.8 Hz is placed in each plot to indicate the location of the high Q peak in the room transfer function.

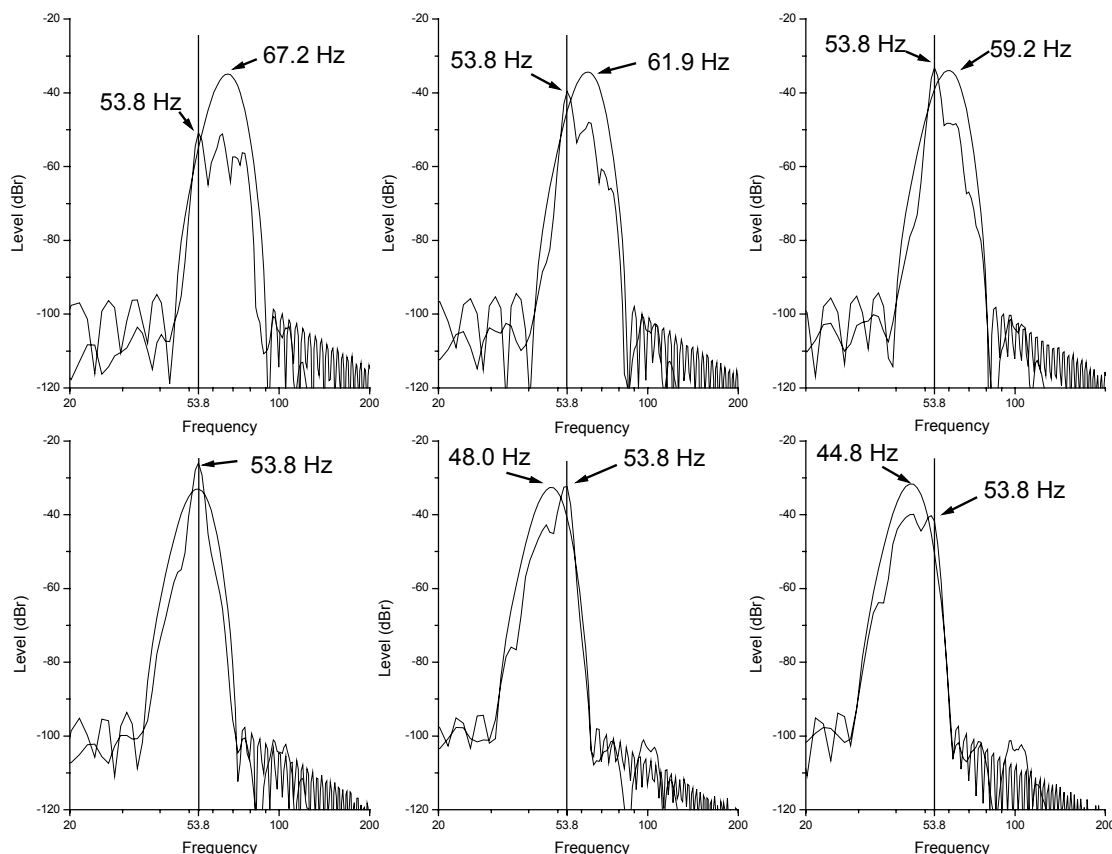


Figure 26: Spectra of Bursts Before and After Convolution with a Room Impulse Response

The room has a profound effect on the spectrum of the original burst. As the frequency of the input bursts decreases, the center frequency of the output burst stays at 53.8 Hz. The dominant frequency at the output continues to be 53.8 Hz until an input frequency of 44.8 Hz is reached, at which point the spectrum of the output burst begins to shift away from that of the peak in the room response. Each convolved burst sounds very similar in pitch to the others, but the first and last burst sound dissonant. The pitches of the original bursts are all distinctly different from each other.

V. LEVEL SETTING OF SUBWOOFER

There are some several millions of audio systems using subwoofers to reproduce low frequencies. The users of those systems usually have no means to optimally adjust the subwoofer level, either by objective means or subjectively. It is expected that a typical user will have set the level of the subwoofer by playing program material and then adjusting the subwoofer level until the subjective balance between low frequencies and the rest of the audio spectrum is pleasing. What is not known is how flat the objectively measured response will be after such a calibration.

It seems apparent that the program material selected as a reference for subjectively setting the subwoofer level must have a “correct” spectral balance. If it does not, presumably the listener will adjust the subwoofer level to make the spectrum correct [18], or at least within the limits of spectral correction by adjustment of the subwoofer level.

If a subwoofer is used in conjunction with other speakers to reproduce the bass frequencies, we assume that it should be set to reproduce at the correct level. By “correct” level, we mean that the objectively measured response of the combined system should be uniform. We also mean that the low frequency response of the combined system should sound like that of a full-range system with uniform low frequency response. These two definitions may not be the same.

Since the room response at low frequencies is extremely non-uniform, there is an uncertainty as to what the “correct” objectively measured response should be. If the low-frequency room transfer function is characterized by having one resonance that is significantly more energetic than the others, then program material having spectral content at those frequencies will have those frequencies emphasized with respect to the rest of the spectrum. If the program contains no components at the frequency of the resonant band, then one subwoofer level is correct. If the program contains only the resonant band, then the correct level is another, lower setting. For broad-band program there is no single correct level. All settings are incorrect; some are better than are others. If the acoustical transfer function cannot be corrected then a compromise subwoofer level setting must be chosen.

For the reasons discussed above it is difficult to properly set the subwoofer level in an audio reproduction system. The calibration can be performed either objectively, using instrumentation, or subjectively, using either program material or special test signals. In order to determine whether an objective or subjective calibration will provide an optimum result an experiment was performed to compare the results of objective calibrations and subjective calibrations, and to ascertain whether the results obtained from the calibrations would be found to be satisfactory. Several experiments were conducted to observe the behavior of listeners when they were asked to subjectively judge the balance of low frequencies to mid and high frequencies and to correct the response by adjusting the subwoofer level. If the setting of the subwoofer level is to be performed subjectively, there are several possible choices for a calibration signal. Calibration can either be performed with program material or with a test signal.

The level-setting experiments were designed to explore the following questions:

1. How do levels set subjectively with program material compare with objectively measured responses?
2. How do levels set subjectively with test signals compare with objectively measured responses?
3. Based on the results from 1 and 2, is one test signal better than the others?
4. If the system is set up objectively using a noise test signal, is it subjectively satisfactory?
5. If the room response is significantly non-flat, how does this affect the level setting?

Test Set-up for Subjective Level-Setting Experiment

The subjective level-setting exercise is divided into two main tasks depending on the type of program material used:

1. Adjusting the subwoofer level to achieve a result that is most nearly “correct” (best spectral balance)
2. Adjusting the subwoofer level to achieve closest match in subjective loudness between two alternating signals

In the first test the listener is presented with full bandwidth program material and asked to set the level so that the spectral balance between the full range speakers and the subwoofer is “correct”. The second test is a comparison between two band-limited signals that are present in either the subwoofer or the main left and right speakers, alternating between the two at 2 second intervals.

Test Environment

The test subwoofer was located in listening room “F”, as described in section III, Figure 16. The listener and subwoofer were located at a position predicted to give the most nearly uniform response. The response at the listening position was verified by objective measurements performed by the same methods as used in part III, and the subwoofer and the listener position were adjusted until the low frequency response was the most uniform that could be achieved. The position of the listener and the subwoofer are the ones marked “2” in Figure 16.

Note that the listener is seated forward relative to the reference listening position, the position at which the listener is equidistant from all five loudspeakers. This position clearly is not optimal for normal listening since it places the listener too far from the surround loudspeakers, and too close to the front loudspeakers.

Selection of Listeners

Six test subjects were selected from individuals with extensive experience in high-fidelity sound reproduction and mixing or formal training in critical listening. All of the listeners were male and between the ages of 25 and 35. None of the listeners reported having had any known hearing pathology. Training was performed to ensure that all subjects could correctly repeat the results of a low frequency level setting exercise.

Performance in subjective listening tests is known to improve with training. It has been shown that there is a correlation between listener selection and variance of results in subjective evaluation. It has been reported [19] that the use of highly experienced listeners can reduce the number of listener required to produce a certain confidence interval by as much as a factor of seven relative to using inexperienced listeners.

Training

The purpose of the training exercise was to give the listeners experience in hearing what the band of frequencies reproduced by the subwoofer sounded like, and to assess the ability of the listeners to perform this task correctly and repeatably. A level-matching test was substituted for the level-setting task to allow the listeners to focus on the subwoofer frequency range. In the training session the listener was provided with an A-B switch which put either a fixed attenuator or a variable attenuator in the path to the subwoofer. The fixed attenuator was set such that objective frequency response was flat throughout the range from the low end of the main speakers and the subwoofer region. When the variable attenuator was adjusted the frequencies in the subwoofer range were either boosted or cut.

The following program material was used for the training exercise:

- Blade Runner, "Blade Runner", (Tr. 1, 1:20 - 2:20, East West, 4509-99675-2)
- Knockin' on Heaven's Door, "Reggae Party", (Tr. 13, 3:22 - 4:22, Island Records, 314-565- 654-2)

The block diagram for the system used in the training exercise is shown in the figure below:

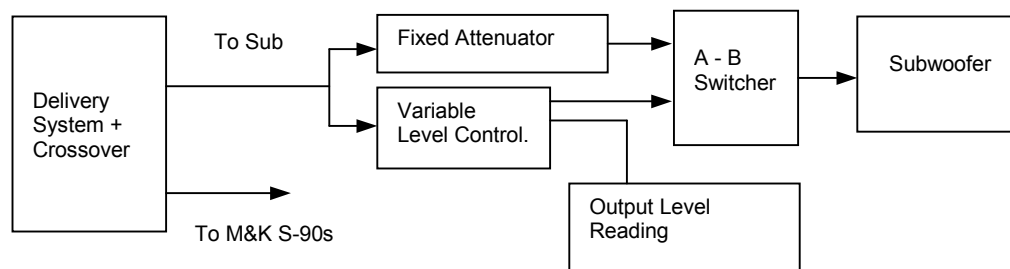


Figure 27: Block Diagram of System Used in Level-Matching (training) Exercise.

Setup and Methodology for Level-Setting Listening Tests

The basic listening test setup was comprised of a multichannel reproduction system using five M&K S-90 main speakers with the low frequencies crossed over to the subwoofer at 80 Hz using a 4th order Butterworth-squared (Linkwitz-Riley) crossover. The subwoofer was the same Velodyne HGS-12 described in earlier sections of this paper. The subwoofer level was adjustable using a continuously variable wired remote level control with a remote readout allowing the experimenter to read the attenuation set to within a resolution of 0.01 dB. The test setup is shown in Figure 28. below:

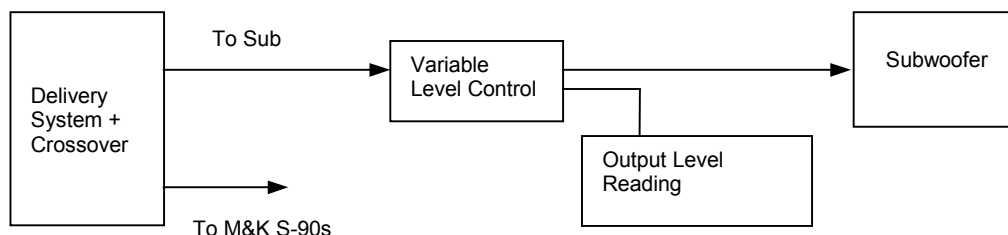


Figure 28. Block Diagram of System Used in Level-Setting Exercise

After the training exercise, the listeners were asked to set the subwoofer level such that the balance of the bass frequencies with respect to the rest of the audio spectrum was correct, or most nearly correct. The subwoofer was capable of being turned down to zero output, or being turned up to a level greater than was chosen by any of the listeners, with the exception of one. The listener who consistently set the subwoofer level higher than the system could provide was eliminated from the tests.

Three 2-channel recordings of popular music were used in the section of the experiment. They were preselected as having a wide and uniform level of LF signal. The three programs were:

Program A – Tracy Chapman, “Fast Car”, (Tracy Chapman, Tr. 2, Electra Records 60774-2/4)

Program B – Bob Marley and the Wailers, “Natural Mystic” (Natural Mystic, Tr. 1, Island Records, 314-524-102 2)

Program C – Jennifer Warnes, “The Hunter” (The Hunter, Tr. 9, BMG 01005-82089-2)

Program D – Pink Noise rolled off below 40 Hz and above 800 Hz to match program spectrum

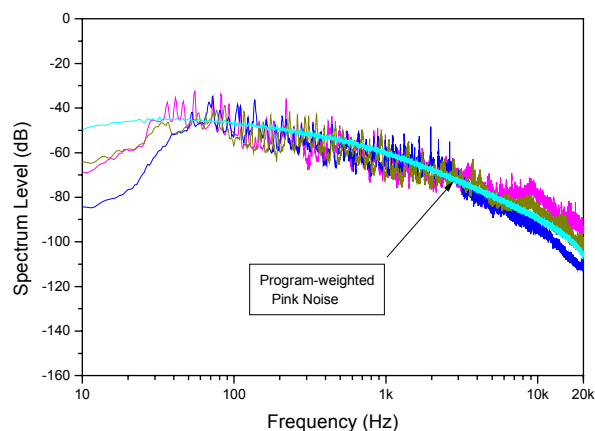


Figure 29. Spectrum of Three Program Segments and Program-Weighted Pink Noise

The spectra of the three program segments and of the program-weighted pink noise can be seen to be very similar, except that the program-weighted noise does not roll off as rapidly below 40 Hz as does the program.

The same level-setting experiment was also performed, using the same listeners and equipment, with combinations of five artificial test signals. The test signals were designed to allow the user to set the subwoofer level using either objective means (sound level meter, real time analyzer) or subjectively, by ear. The five test signals were:

- a. Broad band-pass calibration noise centered at 700 Hz (CN)
- b. 40-80Hz band-pass noise (LBPB)
- c. 80-160 Hz band-pass noise (HBPN)
- d. Pink Noise (PN)
- e. Pink Noise rolled off above 800 Hz and below 40 Hz at 6 dB/octave – Program Weighted Pink Noise (PWPB)

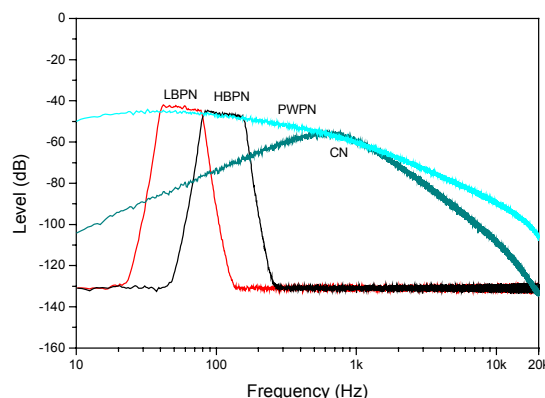


Figure 30. Spectra of Artificial Calibration Signals

The specific groupings or pairings of test signals were:

- A. Broad band-pass calibration noise (CN) vs. 40-80 Hz band-pass noise (LBPB), alternating at 2 second intervals
- B. 80-160 Hz band-pass noise (HBPN) vs. 40-80 Hz band-pass noise (LBPB), alternating at 2 second intervals
- C. Pink noise alone (PN)
- D. Program-weighted Pink noise alone (PWPB)

The intent of pairing A is to give the listener noise signals which are largely constrained to the respective pass-bands of the main loudspeakers or to the subwoofer. Because the contributions of the spectra for each of the test signals to the overall perceived loudness is complex, and in particular is not well understood for very low frequencies, it was not expected that the loudness of the two signals would be the same. Signals that are more similar to each other, and closer in frequency, were created for comparison case B. This is similar to the recommendation of the Surround Sound Forum [25], [26]. In this case the differences in perceived loudness should be less since the rms level is the same and the signals are separated minimally in frequency. The requirement again was either to balance the level of the two test signals (A and B), or to select a subwoofer level that gave the best overall spectral balance (C and D).

Program Level

The threshold of hearing is high and equal loudness contours are closely spaced at low frequencies [20]. The implication of these factors is that it is expected that listener preferences will be strongly affected by playback level. In order to reduce the effect of program level on listener performance, the levels were normalized in the source media. The level of all of the programs was normalized by first measuring the rms A-weighted level (C-weighted in the case of the LBPB and HBPN) of each of the program segments, and then applying a level adjustment to each program segment to make it have a level of -31 dBFS. During reproduction in the listening room, the gain in the reproduction system was adjusted such that the normalized level was reproduced at 69 dB SPL. There is insufficient data on listener preference for sound pressure levels during actual use in the home. A study was conducted on listener preference for television listening level, and for home theatre listening level [21]. In that study the mean preferred dialog level for television listening was 60.5 dB SPL, and for home theatre listening it was 69 dB SPL, A-weighted. Although most of the program segments used were musical material rather than movie material, the home theatre listening level was chosen in the absence of any other firm recommendation.

Results for Level-Setting Listening Test

The training section of the level matching experiment showed that all of the listeners were able to match the adjustable subwoofer level to the fixed (flat frequency response) subwoofer level to within a very tight tolerance. The results of the level matching experiment are shown in the figure below:

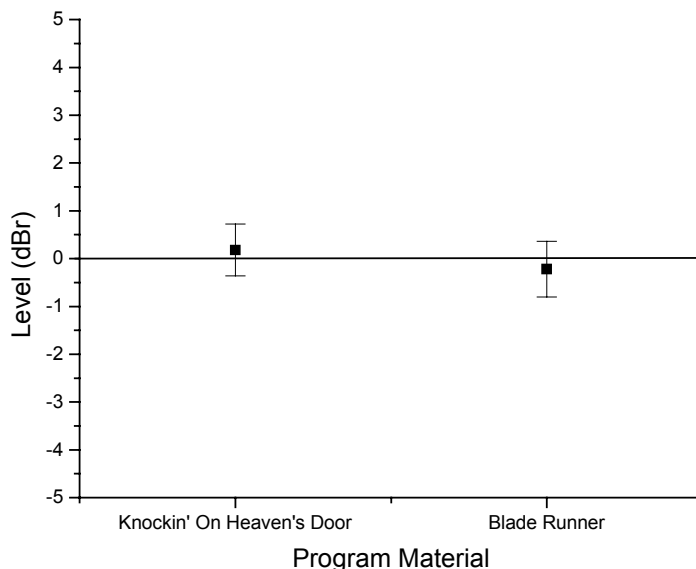


Figure 31. Level Matching of Subwoofer Level with Respect to a Fixed Reference for Program Material

Typically, listeners were able to accomplish the level matching to within an accuracy of ± 0.63 dB within one standard deviation. This implies that the Just Noticeable Difference (JND) for the level of low frequency program material is on the same order.

The results from the subwoofer level setting using program material were different. Individual listeners using a specific program were able to repeatably set the subwoofer level within about a 2 dB interval, with 95% confidence intervals. But the level chosen was different for different programs. In general, for a single listener a different subwoofer level was chosen for different programs, with the difference being statistically significant. It can be concluded that the listener's preference was for different subwoofer levels for different programs. The ordering of the preferred level for the different programs was different for the different listeners. One listener may prefer program A to have a higher subwoofer level than program B, and another listener may the opposite.

It can be concluded from this result alone that it is not possible to set the system to a subwoofer level that would have been within the preference of all listeners, or even for two of the listeners, for all three programs. The authors had originally hypothesized that the listeners would apply a correction to the level of a program that might have had insufficient bass by setting the subwoofer higher, and by attenuating the subwoofer for programs with too much bass. We conclude that there are individual preferences having to do with the specific program that have a stronger effect than the average spectral content of the program.

The results using the various artificial test signals are equally mixed. It is difficult to find a correlation, either between the settings obtained by the various listeners for a given test signal, or between the various test signals, or with the subwoofer level settings obtained using program material.

The results of these tests show that none of the proposed subwoofer calibration signals can be used to set the subwoofer level such that it will be within the preference range of all of the listeners. It usually will not be possible to use one of the subwoofer level-setting signals to set the level so that it will be within the preference range for program material for the same listener.

When the mean results are analyzed the outcome is slightly more favorable.

The mean preferred subwoofer levels for all of the listeners are shown for each of the programs in the figure below:

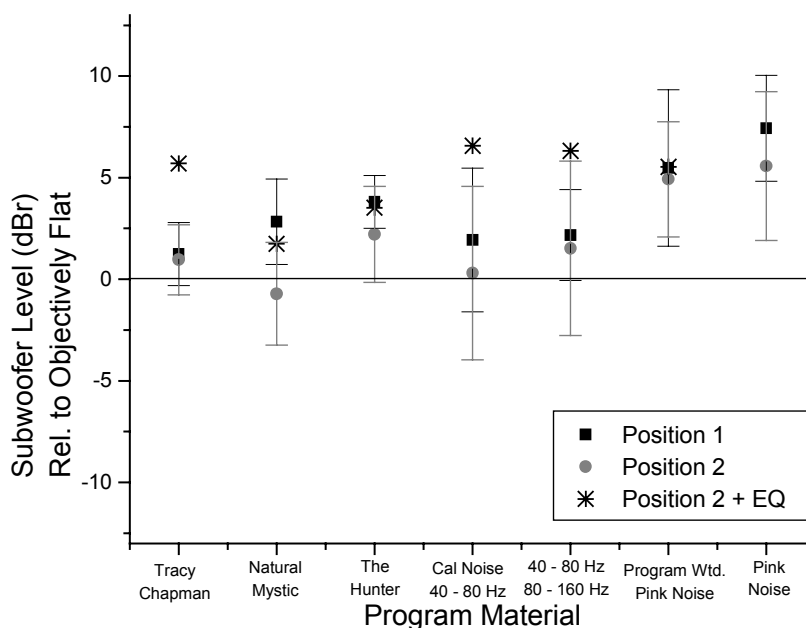


Figure 32. Relative Levels in Level-Setting Listening Test Using Program Material

Position 1 is the position optimized for flattest low-frequency response, while position 2 was selected as having a response which is substantially less flat than for 1. Several observations can be made. The listeners usually set the subwoofer level slightly higher than the objectively flat. The mean levels for the three musical program sources were in a range from -1 dB to $+2$ dB relative to objectively flat response. The total range of subwoofer levels was from -3 dB to $+5$ dB relative to objectively flat response.

The mean subwoofer levels achieved with level-matching and using the two artificial test signals were very similar to what was achieved with program material. The mean subwoofer level using the comparison of “cal noise” and 40-80 Hz band-pass noise was $+0.5$ dB, and the mean subwoofer level using the comparison of 80-160 Hz band-pass noise with 40-80 Hz band-pass noise was $+2$ dB. The over-all range of levels set was slightly greater using the calibration signals than using program material.

The last two level-setting experiments were performed using pink noise and the program weighted pink noise. The program-weighted pink noise signal was created after preliminary tests that indicated that the subwoofer level set using pink noise was too high. Since pink noise has a generally brighter spectrum than does typical program material, it was felt that perhaps the listener was attempting to correct the too-bright spectrum by turning up the subwoofer. The program-weighted pink noise was an attempt to correct this problem, but in the results from the figure it can be seen that both the mean and the total range of subwoofer level settings are not significantly different for the two signals.

A very limited number of trials were conducted with the subwoofer in position 2 (non-flat objective response) and parametric equalization applied, as determined by the algorithm defined in section IV. The equalization approach causes the average level for broad-band program material to drop, since the peaks in the transfer function are reduced, but the dips and notches are left uncorrected. There is a tendency for the levels set for the equalized room transfer function to be slightly higher, but not enough listeners were tested to calculate meaningful statistics.

The process of subwoofer level calibration using program material is surprisingly accurate. Different program material and different listeners will result in a different setting of the subwoofer, and presumably non-expert listeners will not set the subwoofer as accurately. Correctly setting the subwoofer level using program material is predicated on having program material with a “correct” spectral balance. Pink noise and program-weighted pink noise were tried in an effort to create standardized signals for subjective level setting, but have proven to give incorrect results. The two level-matching methods gave equally effective results and may be good candidates for application as standard test signals within products and on test discs.

VI. CONCLUSIONS

The process of surveying the room acoustics in a finite number of installations cannot produce conclusions that are necessarily valid for all such installations. Nonetheless we have observed a number of similarities between those installations that should be noted:

About layout:

- Multichannel loudspeaker layouts are usually narrower than recommended by BS 775.
- The listener is usually located close to the center of the room, exposing him to the maximum deviations due to axial room modes.
- The subwoofer is usually located near the periphery of the room, causing it to be maximally coupled into room modes.

About low-frequency room acoustics:

- Room transfer functions typically vary by about 30 dB through the subwoofer range
- There is a rise in the room transfer function at frequencies below about 30 Hz. Subwoofers can be designed with a compensatory roll-off and achieve flat room response to about 10 Hz.
- The envelope of the spatial distribution varies more than expected.
- There is very often one room mode with an exceptionally large amplitude relative to the others.
- Open doors cause a rise in the extreme low frequencies.
- Domestic rooms usually have lower Q resonances than professional rooms because they have windows and less rigid walls which increase low-frequency absorption.
- Notches in the room response vary considerably in amplitude and frequency throughout the space around the listening position.
- Peaks are usually, but not always, stable in amplitude and frequency throughout the listening space.
- Peaks at one position within the listening space will occasionally turn into a notch at another position.

About room equalization:

- It is usually possible to apply equalization to the largest peak in the room transfer function and reduce its Q at the listening position...
- But in some rooms the largest peak exhibits significant spatial variation, leading to the conclusion that perhaps it shouldn't be equalized, or not to the extent that it might otherwise be.
- Placement equalization can improve the response at the reference listening position but does not improve the distribution.
- Appropriately designed parametric equalization can change the transfer function such that the bass sounds more like the program material does without the effect of the room acoustics.

About the perceptual effect of low frequency room acoustics:

- The high Q of room modes increases the loudness of some notes relative to others.
- High Q room modes filter broad-band audio signals with the effect of causing an audible pitch-shift.
- Interference between modes causes echo-like behavior with a time scale of 50 ms to 400 ms.

About setting the subwoofer level using program material or test signals:

- Individual listeners can use program material to set the subwoofer level to within a very tight tolerance.
- Different program material causes the listener to set the subwoofer to different levels and different listeners set the subwoofer to different levels using the same program material.
- Calibration of the subwoofer level subjectively using program material can give a result, which is close to an objectively flat response.
- Calibration by matching the level of an octave band-pass noise signal to a mid-band noise signal, or by comparing the loudness of two band-pass signals adjacent to the crossover frequency is equally effective.

About performing listening tests comparing subwoofers:

- Moving either the subwoofer or listener will drastically affect the low frequency acoustics.
- When comparing two subwoofers, unless they are in exact same place, the audible difference may be entirely due to variations in the room transfer function and not to the subwoofers themselves.

VII. FUTURE WORK

Several areas addressed in this paper merit additional work. The authors have demonstrated that the resonances caused by modes in the room response, result in audible echoes and pitch shift. Simple parametric equalization reduces but does not eliminate these effects. A better algorithm for the selection of equalization parameters may produce a better subjective result.

The method shown for evaluating the variation in low frequency response in the volume adjacent to the reference listening position is an effective means for deciding which response aberrations can be equalized and which should not, but a more automated means for making that judgement would be useful.

The response at points immediately adjacent to each other is quite similar, and positions further away have larger differences. The authors' data would allow the calculation of the error as a function of distance from the reference listening position, but that hasn't been done.

It has been shown that setting the subwoofer level subjectively using either program material is moderately effective, but it is not known what the acceptance ranges of individual listeners are.

VIII. APOLOGIA AND ACKNOWLEDGEMENTS

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